GSB518 Notes

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Preface

This is a selection of Handouts for Cal Poly GSB 518, Essential Statistics for Business Analytics, covering topics in Probability and Statistics.

1 Randomness and Probability

Probability comes up in a wide variety of situations. Consider just a few examples.

1. The probability that you roll doubles in a turn of a board game is 0.167.
2. The probability that a randomly selected Cal Poly student is a California resident is 0.84
3. The probability that the high temperature in San Luis Obispo, CA tomorrow is above 90 degrees F is 0.4
4. The probability that the Los Angeles Dodgers win the next World Series is 0.27.
5. The probability that extraterrestrial life currently exists somewhere in the universe.
6. The probability that you ate an apple on April 17, 2019.

Example 1.1. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of "probability" the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur.
- Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- Many phenomena involve physical randomness, like flipping a coin or drawing powerballs at random from a bin, or in statistical applications of random sampling or random assignment.
- But in many other situations, randomness just vaguely reflects uncertainty.
- Random does *not* mean haphazard. In a random phenomenon, while individual outcomes are uncertain, there is a *regular distribution of outcomes over a large number of (hypothetical) repetitions*.
- Also, random does *not* necessarily mean equally likely. In a random phenomenon, certain outcomes or events might be more or less likely than others.

- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event's likelihood or degree of uncertainty. A probability can take any value in the continuous scale from 0% to 100%.
- There are two main interpretations of probability.
  - **Long run relative frequency.** The probability of an event can be interpreted as the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon.
  - **Subjective probability.** There are many situations where the outcome is uncertain, but it does not make sense to consider the situation as repeatable. In such situations, a subjective (a.k.a., personal) probability describes the degree of likelihood a given individual ascribes to a certain event. Think of subjective probabilities as measuring relative degrees of likelihood rather than long run relative frequencies.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation. In either case, the same basic logical consistency requirements must be satisfied.
- A **simulation** involves an artificial recreation of the random phenomenon, usually using a computer. The probability of an event can be approximated by simulating the random phenomenon a large number of times and determining the proportion of simulated repetitions on which the event occurred out of the total number of repetitions in the simulation.

Example 1.2. One of the oldest documented problems in probability is the following: If three fair six-sided dice are rolled, what is more likely: a sum of 9 or a sum of 10?

1. Conduct a *simulation* to investigate this question.

2. Use the simulation results to approximate the probability that the sum is 9; repeat for a sum of 10.

3. It can be shown that the theoretical probability that the sum is 9 is $25/216 = 0.116$. Write a clearly worded sentence interpreting this probability as a long run relative frequency.

4. It can be shown that the theoretical probability that the sum is 10 is $27/216 = 0.125$. How many times more likely is a sum of 10 than a sum of 9?

- A probability can be interpreted as a theoretical long run relative frequency.
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency.* For example, if $N = 10000$ then the margin of error is roughly $1/\sqrt{10000} = 0.01$. (But be careful when approximating *conditional* probabilities.)

Example 1.3. What is your subjective probability that Professor Ross has ever attended a "famous" sporting event (like the Superbowl, a World Series game 7, a World Cup match, an NCAA tournament game with a famous shot, an Olympic final, etc)? Consider the following two bets, and suppose you must choose only one.

A. You win \$100 if Professor Ross has attended a famous sporting event, and you win nothing otherwise.
B. A box contains 40 green and 60 gold marbles that are otherwise identical. The marbles are thoroughly mixed and one marble is selected at random. You win \$100 if the selected marble is green, and you win nothing otherwise.

1. Which of the above bets would you prefer? Or are you completely indifferent? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

2. If you preferred bet B to bet A, consider bet C which has a similar setup to B but now there are 20 green and 80 gold marbles. Do you prefer bet A or bet C? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

3. If you preferred bet A to bet B, consider bet D which has a similar setup to B but now there are 60 green and 40 gold marbles. Do you prefer bet A or bet D? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

4. Continue to consider different numbers of green and gold marbles. Can you zero in on your subjective probability?

Example 1.4. Suppose your subjective probabilities for who will be the next World Series champion satisfy the following conditions.

- The Phillies and Mets are equally likely to win
- The Yankees are 1.5 times more likely than the Phillies to win
- The Dodgers are 2 times more likely than the Yankees to win
- The winner is as likely to be among these four teams—Phillies, Mets, Yankees, Dodgers—as not

1. Compute the subjective probability that each team wins.

2. Represent these probabilities in a spinner, like from a kids game.

3. Compute the probability that the winner is not the Dodgers.

4. How many times more likely are the Dodgers to not win than to win? (This is referred to as the "odds".)

5. Compute the probability that the winner is the Dodgers or the Yankees.

- We will use the two interpretations — long run relative frequencies and subjective probabilities — interchangeably.
- With subjective probabilities it is often helpful to consider what might happen in a simulation.
- It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.
- A probability takes a value in the sliding scale from 0 to 100%.
- Don't just focus on computation; always remember to properly interpret probabilities.

Example 1.5. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Flip a coin *which is known to be fair* 10 times.

    a. The probability that the results are, in order, HHHHHHHHHH.
    b. The probability that the results are, in order, HHTHTTTHHT.

2. Flip a coin which is known to be fair 10 times.

    a. The probability that all 10 flips land on H.
    b. The probability that exactly 5 flips land on H.

- Warning! Your psychological judgment of probabilities is often inconsistent with the mathematical logic of probabilities.
- When interpreting probabilities, consider the conditions under which the probabilities were computed, in the proper direction

- When interpreting probabilities, be careful not to confuse "the particular" with "the general".
  - **"The particular:"** A very specific event, surprising or not, often has low probability.
  - **"The general:"** While a very specific event often has low probability, if there are many like events their combined probability can be high.
- Even if an event has extremely small probability, given enough repetitions of the random phenomenon, the probability that the event occurs on *at least one* of the repetitions is often high.

1.1 Exercises

Exercise 1.1. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Consider a Cal Poly student who frequently has blurry, bloodshot eyes, generally exhibits slow reaction time, always seems to have the munchies, and disappears at 4:20 each day. Which of the following events, A or B, has a *higher* probability? (Assume the two probabilities are not equal.)

   a. The student has a GPA above 3.0.
   b. The student has a GPA above 3.0 and smokes marijuana regularly.

2. Randomly select a man.

   a. The probability that a randomly selected man is greater than six feet tall.
   b. The probability that a randomly selected man who plays in the NBA is greater than six feet tall.

3. In the Powerball lottery there are roughly 292 million possible winning number combinations, all equally likely.

   a. The probability you win the next Powerball lottery if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23
   b. The probability you win the next Powerball lottery if you purchase a single ticket, 1-2-3-4-5, plus the Powerball number, 6.

4. Continuing with the Powerball

   a. The probability that the numbers in the winning number are not in sequence (e.g., 4-8-15-16-42-23)
   b. The probability that the numbers in the winning number are in sequence (e.g., 1-2-3-4-5-6)

5. Continuing with the Powerball
   a. The probability that you win the next Powerball lottery if you purchase a single ticket.
   b. The probability that someone wins the next Powerball lottery. (FYI: especially when the jackpot is large, there are hundreds of millions of tickets sold.)

Exercise 1.2. Your favorite local weatherperson forecasts a 30% chance of rain tomorrow and a 60% chance of rain the next day in your city.

1. Explain how these probabilities are subjective.

2. You ask Donny Don't to interpret the 30% as a long run relative frequency. Donny says: "it will rain in 30% of the city tomorrow". You ask him to elaborate; he says: "Well, there are many different locations in the city. In some of the locations it will rain, in some it won't. It will rain in 30% of the locations, and not in the other 70%. That is, rain will cover 30% of the area of the city, and the other 70% won't have rain." Do you agree? If not, how would you interpret the 30% as a long run relative frequency?

3. You ask Donny Don't to interpret the values 30% and 60% as relative degrees of likelihood. Donny says: "Well, 30% is not that big, so it's not going to rain that hard tomorrow. Also, 60% is twice is big as 30%, so it's going to rain twice as hard two days from now as it will tomorrow". Do you agree? If not, how would you interpret the 30% and 60% as relative degrees of likelihood?

4. Donny says: "Can't we just look at the data from all the days with weather conditions similar to the ones forecast for tomorrow, and see how often it rained on those days to find the probability of rain tomorrow? No subjectivity about that!" How would you respond?

- A **probabilistic forecast** combines observed data and statistical or mathematical models to make predictions.
- Rather than providing a single prediction such as "it will rain tomorrow", probabilistic forecasts provide a range of scenarios and their relative likelihoods.
- Such forecasts are subjective in nature, relying upon the data used and assumptions of the model.
- Changing the data or assumptions can result in different forecasts and probabilities.

2 Probability Models

- Due to the wide variety of types of random phenomena, an **outcome** can be virtually anything. In particular, an outcome does *not* have to be a number.
- The **sample space** is the set of all possible outcomes of the random phenomenon..
- An event is something that could happen. That is, an *event* is a collection of outcomes that satisfy some criteria. If the sample space outcomes are represented by rows in a spreadsheet, then an event is a subset of rows that satisfies some criteria
- Events are typically denoted with capital letters near the start of the alphabet, with or without subscripts (e.g. A, B, C, A_1, A_2). Events can be composed from others using basic set operations like unions ($A \cup B$), intersections ($A \cap B$), and complements (A^c).
  - Read A^c as "not A".
  - Read $A \cap B$ as "A and B"
  - Read $A \cup B$ as "A or B". Note that unions ($\cup$, "or") are always inclusive. $A \cup B$ occurs if A occurs but B does not, B occurs but A does not, or both A and B occur.
- A collection of events $A_1, A_2, \ldots$ are **disjoint** (a.k.a. mutually exclusive) if $A_i \cap A_j = \emptyset$ for all $i \neq j$. That is, multiple events are disjoint if none of the events have any outcomes in common.
- A **probability measure**, typically denoted P, assigns probabilities to *events* to quantify their relative likelihoods according to the assumptions of the model of the random phenomenon.
- The probability of event A, computed according to probability measure $\mathrm{P}(A)$, is denoted $\mathrm{P}(A)$.
- A valid probability measure P must satisfy the following three logical consistency "axioms".
  - For any event A, $0 \leq \mathrm{P}(A) \leq 1$.
  - If Ω represents the sample space then $\mathrm{P}(\Omega) = 1$.
  - (*Countable additivity.*) If $A_1, A_2, A_3, \ldots$ are disjoint then

$$\mathrm{P}(A_1 \cup A_2 \cup A_2 \cup \cdots) = \mathrm{P}(A_1) + \mathrm{P}(A_2) + \mathrm{P}(A_3) + \cdots$$

- Additional properties of a probability measure follow from the axioms
  - *Complement rule.* For any event A, $\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$.
  - *Subset rule.* If $A \subseteq B$ then $\mathrm{P}(A) \leq \mathrm{P}(B)$.
  - *Addition rule for two events.* If A and B are any two events

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

  - *Law of total probability.* If $C_1, C_2, C_3 \ldots$ are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \cdots = \Omega$, then
$$\mathrm{P}(A) = \mathrm{P}(A \cap C_1) + \mathrm{P}(A \cap C_2) + \mathrm{P}(A \cap C_3) + \cdots$$

- The axioms of a probability measure are minimal logical consistent requirements that ensure that probabilities of different events fit together in a valid, coherent way.
- A single probability measure corresponds to a particular set of assumptions about the random phenomenon.

Example 2.1. Roll a four-sided die twice, and record the result of each roll in sequence. For example, a 3 on the first roll and a 1 on the second is not the same outcome as a 1 on the first roll and a 3 on the second.

1. Identify the sample space.

2. We might be interested in the sum of the two rolls. Explain why it is still advantageous to define the sample space as in the previous part, rather than as just $2, \ldots, 8$.

Example 2.2. Regina and Cady are meeting for lunch. They will each definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval [0, 60].

Draw a picture to represent the sample space.

Example 2.3. Roll a four-sided die twice, and record the result of each roll in sequence. Using the sample space from Example 2.1, identify the following events.

1. A, the event that the sum of the two dice is at most 4.

2. B, the event that the larger of the two rolls (or the common roll if a tie) is 3.

3. B^c (identify and interpret).

4. $A \cap B$ (identify and interpret).

5. C, the event that the first roll is a 3.

Example 2.4. Regina and Cady are meeting for lunch. They will each definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval [0, 60].

Using the sample space from Example 2.2, draw a picture to represent each of the following events.

1. A, the event that Regina arrives after Cady.

2. B, the event that either Regina or Cady arrives before 12:30.

3. C, the event that they arrive within 15 minutes of each other.

4. D, the event that Regina arrives before 12:24.

- For a sample space Ω with finitely many possible outcomes, assuming **equally likely outcomes** corresponds to a probabiliy measure P which satisfies

$$\mathrm{P}(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} \qquad \text{when outcomes are equally likely}$$

- Note that equally likely outcomes is an *assumption*, and by no means a requirement. In most situations equally likely outcomes is *not* a reasonable assumption.

Example 2.5. Roll a fair four-sided die twice, and record the result of each roll in sequence. Let P represent the probability measure which assumes equally likely outcomes.

1. How many possible outcomes are there? Why is it reasonable to assume they are equally likely?

2. Compute $\mathrm{P}(A)$, where A is the event that the sum of the two dice is at most 4.

3. Compute $\mathrm{P}(B)$, where B the event that the larger of the two rolls (or the common roll if a tie) is 3.

4. Compute and interpret $\mathrm{P}(A \cap B)$. (Is it equal to $\mathrm{P}(A)\mathrm{P}(B)$?)

5. Compute $\mathrm{P}(C)$, where C the event that the first roll is a 3.

- The continuous analog of equally likely outcomes is a **uniform probability measure**. When the sample space is uncountable, size is measured continuously (length, area, volume) rather that discretely (counting).

$$\mathrm{P}(A) = \frac{|A|}{|\Omega|} = \frac{\text{size of } A}{\text{size of } \Omega} \qquad \text{if P is a uniform probability measure}$$

- Note that a uniform probability measure is an *assumption*, and by no means a requirement. In most situations a uniform probability measure is *not* a reasonable assumption.

Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60.

Recall the sample space from Example 2.2 and assume a uniform probability measure P.

1. Find the probability that Regina arrives after Cady.

2. Find the probability that either Regina or Cady arrives before 12:30.

3. Find the probability that Cady arrives first and Regina arrives at most 15 minutes after Cady.

4. Find the probability that Regina arrives before 12:24.

2.1 Exercises

Exercise 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. (These values are based on the 2018 General Social Survey (GSS).)

1. Represent the information provided using proper symbols.

2. Donny Don't says: "the probability that a randomly selected U.S. household has a pet dog OR a pet cat is $0.47 + 0.25 = 0.72$." Do you agree? What must be true for Donny to be correct? Explain. (Hint: for the remaining parts it helps to consider two-way tables.)

3. What is the *smallest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur.

4. What is the *largest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur. What would be the probability that a randomly selected U.S. household has a pet dog OR a pet cat in this scenario?

5. Donny Don't says: "I remember hearing once that in probability OR means add and AND means multiply. So the probability that a randomly selected U.S. household has a pet dog AND a pet cat is $0.47 \times 0.25 = 0.1175$." Do you agree? Explain.

6. According to the GSS, the probability that a randomly selected U.S. household has a pet dog AND a pet cat is 0.15. Compute the probability that a randomly selected U.S. household has a pet dog OR a pet cat.

7. Compute and interpret $\mathrm{P}(C \cap D^c)$.

Exercise 2.2. Each question on a multiple choice test has four options. You know with certainty the correct answers to 70% of the questions. For 20% of the questions, you can eliminate two of the incorrect choices with certainty, but you guess at random among the remaining two options. For the remaining 10% of questions, you have no idea and guess one of the four options at random.

1. Randomly select a question from this test. What is the probability that you answer the question correctly? (Hint: make a two-way table.)

2. For any given question on the exam, your probability of answering it correctly is either 1, 0.5, or 0.25, depending on if you know it, can eliminate two choices, or are just guessing. How does your probability of correcting answering a randomly selected question relate to these three values? Which value — 1, 0.5, or 0.25 —is the overall probability closest to, and why?

3 Conditional Probability

Example 3.1. The probability[1] that a randomly selected American adult (18+) uses Snapchat is 0.24.

1. Suppose the randomly selected adult is under age 30. Do you think the probability that a randomly selected *adult who is under age 30* uses Snapchat is 0.24? What if the adult is over age 30?

2. The probability[2] that a randomly selected American adult is under age 30 is 0.20. Start to create a two-way table representing 10000 hypothetical U.S. adults. Is the probability that a randomly selected American adult both (1) is under age 30, *and* (2) uses Snapchat equal to 0.20×0.24? Explain.

3. Suppose that the probability that a randomly selected American adult both is use age 30 and uses Snapchat is 0.13. Construct an appropriate two-way table.

4. Find the probability that a randomly selected American *adult who is under age 30* uses Snapchat.

[1] The values in this problem are based on a April, 2021 report by the Pew Research Center.
[2] Based on data from the U.S. Census Bureau

5. How can the probability in the previous part be written in terms of the probabilities provided earlier?

6. Find the probability that a randomly selected American *adult who is over age 30* uses Snapchat.

7. Find the probability that a randomly selected American *adult who uses Snapchat* is under age 30.

- The **conditional probability of event** A given event B, denoted $\mathrm{P}(A|B)$, is defined as

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

- The conditional probability $\mathrm{P}(A|B)$ represents how the likelihood or degree of uncertainty of event A should be updated to reflect information that event B has occurred.
- Conditional probabilities are probabilities. There are analagous conditional versions of the properties of probability, e.g. $\mathrm{P}(A^c|B) = 1 - \mathrm{P}(A|B)$.
- In general, knowing whether or not event B occurs influences the probability of event A. That is,

$$\text{In general, } \mathrm{P}(A|B) \neq \mathrm{P}(A)$$

- Be careful: order is essential in conditioning. That is,

$$\text{In general, } \mathrm{P}(A|B) \neq \mathrm{P}(B|A)$$

- Within the context of two events, we have joint, conditional, and marginal probabilities.
  - Joint: unconditional probability involving both events, $\mathrm{P}(A \cap B)$.
  - Conditional: conditional probability of one event given the other, $\mathrm{P}(A|B)$, $\mathrm{P}(B|A)$.
  - Marginal: unconditional probability of a single event $\mathrm{P}(A)$, $\mathrm{P}(B)$.
- The relationship $\mathrm{P}(A|B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$ can be stated generically as

$$\text{conditional} = \frac{\text{joint}}{\text{marginal}}$$

- When dealing with probabilities (or proportions or percentages) be sure to ask "probability *of what*?" Thinking in fraction terms, be careful to identify the correct reference group which corresponds to the denominator.

Example 3.2. The following two phrases contain exactly the same words, just in different orders. Which is larger, the numerical value of 1 or 2?

1. The probability that a randomly selected man who is greater than six feet tall plays in the National Basketball Association (NBA).
2. The probability that a randomly selected man who plays in the National Basketball Association (NBA) is greater than six feet tall.

Example 3.3. Suppose

- 20% of adults are under age 30
- 63% of adults under age 30 use TikTok
- 31% of adults over age 30 TikTok

1. Use the information to set up a hypothetical two-way table.

2. Compute the probability that an adult uses TikTok.

3. Compute the probability that an adult who uses TikTok is under age 30.

- Rearranging the definition of conditional probability we get the **Multiplication rule:** the probability that two events both occur is

$$\begin{aligned} \mathrm{P}(A \cap B) &= \mathrm{P}(A|B)\mathrm{P}(B) \\ &= \mathrm{P}(B|A)\mathrm{P}(A) \end{aligned}$$

- The multiplication rule says that you should think "multiply" when you see "and". However, be careful about *what* you are multiplying: to find a joint probability you need an unconditional and an appropriate conditional probability.
- You can condition either on A or on B, provided you have the appropriate marginal probability; often, conditioning one way is easier than the other.
- Be careful: the multiplication rule does *not* say that $\mathrm{P}(A \cap B)$ is the same as $\mathrm{P}(A)\mathrm{P}(B)$.
- **Law of total probability** If C_1, C_2, C_3 ... are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \cdots = \Omega$, then

$$\begin{aligned} \mathrm{P}(A) &= \mathrm{P}(A \cap C_1) + \mathrm{P}(A \cap C_2) + \mathrm{P}(A \cap C_2) + \cdots \\ &= \mathrm{P}(A|C_1)\mathrm{P}(C_1) + \mathrm{P}(A|C_2)\mathrm{P}(C_2) + \mathrm{P}(A|C_3)\mathrm{P}(C_3) + \cdots \end{aligned}$$

- The events $C_1, C_2, C_3, \ldots$, which represent the "cases", form a *partition* of the sample space; each outcome $\omega \in \Omega$ lies in exactly one of the C_i.
- The law of total probability says that we can interpret the unconditional probability $\mathrm{P}(A)$ as a probability-weighted average of the case-by-case conditional probabilities $\mathrm{P}(A|C_i)$ where the weights $\mathrm{P}(C_i)$ represent the probability of encountering each case.

3.1 Exercises

Exercise 3.1. Continuing Exercise 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. the probability that a randomly selected U.S. household has a pet dog and a pet cat is 0.15.

In addition to computing, represent the probabilities using appropriate notation.

1. Compute the probability that a randomly selected U.S. household that has a pet dog also has a pet cat. Is it 0.25?

2. Compute the probability that a randomly selected U.S. household that does not have a pet dog has a pet cat. Is it the same as the previous part? Is it one minus the previous part?

3. Describe in words the probability that results from subtracting the answer to the first part from 1.

4. Compute the probability that a randomly selected U.S. household that has a pet cat also has a pet dog.

Exercise 3.2. Suppose that[3]

- 67% of Democrats believe in human-driven climate

[3]Probabilities are estimated based on this 2024 survey.

- 46% of Independents believe in human-driven climate
- 34% of Republicans believe in human-driven climate

Also suppose that[4]

- 28% of American adults are Democrats
- 42% of American adults are Independents
- 30% of American adults are Republicans

1. Define the event A to represent "believes in human-driven climate change" and D, I, R to correspond to affiliation in each of the parties. If the probability measure P corresponds to selecting an American adult uniformly at random, write all the percentages above as probabilities using proper notation.

2. Construct an appropriate two-way table of probabilities.

3. Now suppose that the randomly selected American believes in human-driven climate change. How does this information change the probability that the selected American belongs to each political party? Answer by computing appropriate probabilities (and using proper notation).

4. How many times more likely is it for an American *adult* to believe in human-driven climate change and be Independent than to:

    a. believe in human-driven climate change and be Democrat

[4]Estimate based on Gallup poll

b. believe in human-driven climate change and be Republican

5. How many times more likely is it for an American adult *who believes in human-driven climate change* to be Independent than to be:

    a. Democrat

    b. Republican

6. What do you notice about the answers to the two previous parts?

- The process of conditioning can be thought of as **"slicing and renormalizing".**
    - Extract the "slice" corresponding to the event being conditioned on (and discard the rest). For example, a slice might correspond to a particular row or column of a two-way table.
    - "Renormalize" the values in the slice so that corresponding probabilities add up to 1.
- Slicing determines the *shape*; renormalizing determines the *scale*.
- Slicing determines relative probabilities; renormalizing just makes sure they add up to 1.

Exercise 3.3. Continuing Exercise 3.2. Explain in detail how you could conduct a simulation, using only the information provided in the problem set up, and how you would use the results to approximate the conditional probability of each party given belief in human driven climate change.

- Conditional probabilities can be approximated by filtering (or subsetting) based on the condition
- To approximate $\mathrm{P}(A|B)$, simulate the random phenomenon for a set number of repetitions (say 10000), *discard those repetitions on which* B does not occur, and compute the relative frequency of A among the remaining repetitions (on which B does occur).

- Be careful! The margin of error is based on only the number of repetitions used to compute the relative frequency. So if you perform 10000 repetitions but B occurs only on 2000, then the margin of error for estimate $\mathrm{P}(A|B)$ is roughly on the order of $1/\sqrt{2000} = 0.022$ (rather than $1/\sqrt{10000} = 0.01$. Especially if $\mathrm{P}(B)$ is small, the margin of error could be large resulting in an imprecise estimate of $\mathrm{P}(A|B)$. (Advantage: not computationally intensive.)

4 Bayes Rule

- *Bayes' rule* describes how to update uncertainty in light of new information, evidence, or data.

Example 4.1. There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Suppose that among Antarctic penguins[1] of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that

- 83.4% of Adelie penguins are below average weight,
- 89.7% of Chinstrap penguins are below average weight, and
- 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

1. Use the information to construct an appropriate two-way table of probabilities.

2. Compute the probability that a randomly selected penguin is below average weight. How is this related to the values 0.834, 0.897, 0.049?

3. If the selected penguin is below average weight, what is the probability that it is species A? Species C? Species G?

[1] Throughout "penguin" refers to an Antarctic penguin of one of these three species. This example is based on the famous Palmer penguins data set.

- **Bayes' rule for events** specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$\mathrm{P}(H|E) = \frac{\mathrm{P}(E|H)\mathrm{P}(H)}{\mathrm{P}(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $\mathrm{P}(H)$ is the unconditional or **prior probability** of H (prior to observing evidence E)
- $\mathrm{P}(H|E)$ is the conditional or **posterior probability** of H after observing evidence E.
- $\mathrm{P}(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H

Example 4.2. Continuing Example 4.1. Randomly select a penguin and suppose it is below average weight. We are interested in classifying the species of the penguin.

1. Frame this problem in the context of Bayes rule: Identify the hypotheses, prior probabilities, evidence, likelihood, and posterior probabilities.

2. Prior to observing the penguin's weight, how many times more likely is it to be Adelie than to be Gentoo?

3. How many times more likely is it for a penguin to be below average weight given that it is Adelie than given that it is Gentoo?

4. Compute the posterior probability of each species given that the penguin is below average weight. Given that the penguin is below average weight, how many times more likely is it to be Adelie than to be Gentoo?

5. How are the ratios in the three previous parts related?

6. Apply the same logic to compare species A and C.

- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \ldots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).
- Combining Bayes' rule with the law of total probability,

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)} \end{aligned}$$

$$P(H_j|E) \propto P(E|H_j)P(H_j)$$

- The symbol $\propto$ is read "is proportional to". The relative *ratios* of the posterior probabilities of different hypotheses are determined by the product of the prior probabilities and the likelihoods, $P(E|H_j)P(H_j)$. The marginal probability of the evidence, $P(E)$, in the denominator simply normalizes the numerators to ensure that the updated probabilities sum to 1 over all the distinct hypotheses.
- **In short, Bayes' rule says**

$$\textbf{posterior} \propto \textbf{likelihood} \times \textbf{prior}$$

Example 4.3. Continuing Example 4.2. Randomly select a penguin. We are interested in classifying the species of the penguin depending on whether or not it is below average weight.

1. Before you know the weight of the penguin (or any other information) what specify would you predict it to be? Why? What is the probability that your classification is correct?

2. If the penguin is below average weight, what species would you classify it as? Why?

3. Now suppose that the penguin is not below average weight. Use a Bayes table to compute the posterior probability of each species given that the penguin is not below average weight. If the penguin is not below average weight, what species would you classify it as? Why?

4. Suppose that you classify any randomly selected penguin based on whether or not it is below average weight as in the two previous parts. What is the posterior probability that you classify the penguin correctly? How does this compare to your probability of being correct before you observed the penguin's weight?

- Bayesian analysis is often an iterative process.
- Posterior probabilities are updated after observing some information or data. These probabilities can then be used as prior probabilities before observing new data.
- Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
- The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way. Also, the final posterior probabilities are not impacted by the order in which the data are observed.

4.1 Exercises

Exercise 4.1. A recent survey of American adults asked: "Based on what you have heard or read, which of the following two statements best describes the scientific method?"

- 70% selected "The scientific method produces findings meant to be continually tested and updated over time". (We'll call this the "iterative" opinion.)
- 14% selected "The scientific method identifies unchanging core principles and truths". (We'll call this the "unchanging" opinion).
- 16% were not sure which of the two statements was best.

How does the response to this question change based on education level? Suppose education level is classified as: high school or less (HS), some college but no Bachelor's degree (college), Bachelor's degree (Bachelor's), or postgraduate degree (postgraduate). The education breakdown is

- Among those who agree with "iterative": 31.3% HS, 27.6% college, 22.9% Bachelor's, and 18.2% postgraduate.
- Among those who agree with "unchanging": 38.6% HS, 31.4% college, 19.7% Bachelor's, and 10.3% postgraduate.
- Among those "not sure": 57.3% HS, 27.2% college, 9.7% Bachelor's, and 5.8% postgraduate

1. Consider the conditional probability that a randomly selected American adult agrees that the scientific method is "iterative" given that they have a postgraduate degree. Identify the prior probability, hypothesis, evidence, likelihood, and posterior probability, and use Bayes' rule to compute the posterior probability.

2. Find the conditional probability that a randomly selected American adult with a postgraduate degree agrees that the scientific method is "unchanging".

3. Find the conditional probability that a randomly selected American adult with a postgraduate degree is not sure about which statement is best.

4. How many times more likely is it for an *American adult* to agree with the "iterative" statement than to agree with the "unchanging" statement?

5. How many times more likely is it for an American adult to have a postgraduate degree when the adult agrees with the iterative statement than when the adult agree with the unchanging statement?

6. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the "iterative" statement than to agree with the "unchanging" statement?

7. How are the values in the three previous parts related?

Exercise 4.2. Now suppose we want to compute the posterior probabilities for an American adult's perception of the scientific method given that the randomly selected American adult has some college but no Bachelor's degree ("college").

1. Before computing, make an educated guess for the posterior probabilities. In particular, will the changes from prior to posterior be more or less extreme given the American has some college but no Bachelor's degree than when given the American has a postgraduate degree? Why?

2. Construct a Bayes table and compute the posterior probabilities. Compare to the posterior probabilities given postgraduate degree from the previous examples.

Exercise 4.3. Within both the colleges of Agriculture and Architecture at Cal Poly, about 49% of admitted students are female, about 84% of admitted students went to high school in CA, and the median GPA of admitted students is about 4.1.

An orientation group of 100 newly admitted Cal Poly students includes 75 students in Agriculture and 25 students in Architecture. A student is randomly selected from this group. The selected student is Maddie, who is female, went to high school in CA, and had a high school GPA of 4.1.

Donny Don't says, "The information about Maddie applies equally well to Agriculture or Architecture and doesn't help us decide which college she's in, so it's just 50/50. Given the information about Maddie, the conditional probability that she is in Agriculture is 0.5." Do you agree? If not, what is the conditional probability that Maddie is in the college of Agriculture given the information about her?

5 Independence of Events

Example 5.1. Do Americans think it is important for federal judges to be impartial in how they decide court cases? Consider the following data from the Pew Research Center. (Indepedendents are classified as either lean Democrat or lean Republican.)

	Democrat/lean Dem (D)	Republican/lean Rep (D^c)	Total
Important (I)	1674	1531	3205
Not important (I^c)	146	133	279
Total	1820	1664	3484

Suppose a person is randomly selected from this sample. Consider the events

$$\begin{aligned} I &= \{\text{person thinks it is important for judges to be impartial}\} \\ D &= \{\text{person is a Democrat/leans Dem}\} \end{aligned}$$

1. Compute and interpret $\mathrm{P}(I)$.

2. Compute and interpret $\mathrm{P}(I|D)$.

3. Compute and interpret $\mathrm{P}(I|D^c)$.

4. What do you notice about $\mathrm{P}(I)$, $\mathrm{P}(I|D)$, and $\mathrm{P}(I|D^c)$?

5. Compute and interpret $\mathrm{P}(D)$.

6. Compute and interpret $\mathrm{P}(D|I)$.

7. Compute and interpret $\mathrm{P}(D|I^c)$.

8. What do you notice about $\mathrm{P}(D)$, $\mathrm{P}(D|I)$, and $\mathrm{P}(D|I^c)$?

9. Compute and interpret $\mathrm{P}(D \cap I)$.

10. What is the relationship between $\mathrm{P}(D \cap I)$ and $\mathrm{P}(D)$ and $\mathrm{P}(I)$?

11. When randomly selecting a person from this particular group, would you say that events

D and I are independent? Why?

- Events A and B are **independent** if the knowing whether or not one occurs does not change the probability of the other.
- For events A and B (with $0 < \mathrm{P}(A) < 1$ and $0 < \mathrm{P}(B) < 1$) the following are equivalent. That is, if one is true then they all are true; if one is false, then they all are false.

$$
\begin{aligned}
&A \text{ and } B \text{ are independent}\\
\mathrm{P}(A \cap B) &= \mathrm{P}(A)\mathrm{P}(B)\\
\mathrm{P}(A^c \cap B) &= \mathrm{P}(A^c)\mathrm{P}(B)\\
\mathrm{P}(A \cap B^c) &= \mathrm{P}(A)\mathrm{P}(B^c)\\
\mathrm{P}(A^c \cap B^c) &= \mathrm{P}(A^c)\mathrm{P}(B^c)\\
\mathrm{P}(A|B) &= \mathrm{P}(A)\\
\mathrm{P}(A|B) &= \mathrm{P}(A|B^c)\\
\mathrm{P}(B|A) &= \mathrm{P}(B)\\
\mathrm{P}(B|A) &= \mathrm{P}(B|A^c)
\end{aligned}
$$

Example 5.2. This is a simplified example of *batch testing*[1] for a disease. In batch testing, specimens (like blood or saliva samples) from a group of people are pooled together into one batch, which then undergoes one test. If none of the people has the disease, then the batch test result will be negative, and no further tests are required. But if at least one person in the batch has the disease, then the batch test result will be positive, and then each person in the batch must be tested individually. (We're assuming no false positives and no false negatives for this example.)

Suppose that 8 people are to be tested, first in a single batch. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person. (The assumption of independence might be unreasonable if the people are in close contact with one another.)

1. Identify the possible values for the total number of tests conducted.

[1] Thanks to Allan Rossman for this example. His blog has more investigations into batch testing, including what happens as n and p change. Cal Poly implemented versions of batch testing for COVID through saliva samples and wastewater testing.

2. Compute the probability of each possible value.

- Events $A_1, A_2, A_3, \ldots$ are **independent** if:
  - any pair of events $A_i, A_j, (i \neq j)$ satisfies $\mathrm{P}(A_i \cap A_j) = \mathrm{P}(A_i)\mathrm{P}(A_j)$,
  - and any triple of events A_i, A_j, A_k (distinct i, j, k) satisfies $\mathrm{P}(A_i \cap A_j \cap A_k) = \mathrm{P}(A_i)\mathrm{P}(A_j)\mathrm{P}(A_k)$,
  - and any quadruple of events satisfies $\mathrm{P}(A_i \cap A_j \cap A_k \cap A_m) = \mathrm{P}(A_i)\mathrm{P}(A_j)\mathrm{P}(A_k)\mathrm{P}(A_m)$,
  - and so on.
- Intuitively, a collection of events is independent if knowing whether or not any combination of the events in the collection occur does not change the probability of any other event in the collection.
- When events are independent, the multiplication rule simplifies greatly.

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n) = \mathrm{P}(A_1)\mathrm{P}(A_2)\mathrm{P}(A_3)\cdots\mathrm{P}(A_n) \quad \text{if } A_1, A_2, A_3, \ldots, A_n \text{ are independent}$$

- When a problem involves independence, you will want to take advantage of it. Work with "and" events whenever possible in order to use the multiplication rule. For example, for problems involving "at least one" (an "or" event) take the complement to obtain "none" (an "and" event).

Example 5.3. Continuing Example 5.2. Now suppose that 8 people to be tested are split into two groups of 4 people each. Within each group of 4 people, specimens are combined into a batch to be tested. If anyone in the batch has the disease, then the batch test will be positive, and those 4 people will need to be tested individually. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person.

1. Identify the possible values for the total number of tests conducted.

2. Compute the probability of each possible value.

5.1 Exercises

Exercise 5.1. A certain system consists of four identical components. Suppose that the probability that any particular component fails is 0.1, and failures of the components occur independently of each other. Find the probability that the system fails if:

1. The components are connected in *parallel*: the system fails only if *all* of the components fail.

2. The components are connected in *series*: the system fails whenever *at least one* of the components fails.

3. Donny Don't says the answer to the previous part is $0.1 + 0.1 + 0.1 + 0.1 = 0.4$. Explain the error in Donny's reasoning.

Exercise 5.2. Roll two fair four-sided dice, one green and one gold. There are 16 total possible outcomes (roll on green, roll on gold), all equally likely. Consider the event $E = \{\text{the green die lands on 1}\}$. Answer the following questions by computing and comparing appropriate probabilities.

1. Consider $A = \{\text{the gold die lands on 4}\}$. Are A and E independent?

2. Consider $B = \{\text{the sum of the dice is } 3\}$. Are B and E independent?

3. Consider $C = \{\text{the sum of the dice is } 5\}$. Are C and E independent?

- Independence concerns whether or not the occurrence of one event affects the *probability* of the other.
- Given two events it is not always obvious whether or not they are independent.

- Independence depends on the underlying probability measure. Events that are independent under one probability measure might not be independent under another.
- Independence is often assumed. Whether or not independence is a valid assumption depends on the underlying random phenomenon.

6 Discrete Random Variables

- Roughly, a *random variable* assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- Mathematically, a **random variable (RV)** X is a *function* that takes an outcome in the sample space as input and returns a real number as output
- The random variable itself is typically denoted with a capital letter (X); possible values of that random variable are denoted with lower case letters (x).
  - Think of the capital letter X as a label standing in for a formula like "the number of heads in 4 flips of a coin" and
  - x as a dummy variable standing in for a particular value like 3.
- **Discrete random variables** take at most countably many possible values (e.g., $0, 1, 2, ...$). They are often counting variables (e.g., the number of Heads in 10 coin flips).
- **Continuous random variables** can take any real value in some interval (e.g., $[0, 1]$, $[0, \infty)$, $(-\infty, \infty)$.). That is, continuous random variables can take uncountably many different values. Continuous random variables are often measurement variables (e.g., height, weight, income).
- A function of a random variable is also a random variable: if X is a RV then so is $g(X)$
- Sums and products, etc., of random variables *defined on the same sample space* are random variables. IfX and Y are RVs defined on the same sample space then so are $X + Y$, $X - Y$, XY
- If the sample space outcomes are represented by rows in a spreadsheet, then random variables correspond to columns.
- Expressions like $X = x$ or $\{X = x\}$ represent *events*: for which outcomes is the value of the random variable X equal to the value x. (Remember, if the sample space outcomes are represented by rows in a spreadsheet, then an event corresponds to a subset of rows (outcomes) that satisfies some criteria.)
- The **(probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).

Table 6.1: Two rolls of a fair four-sided die

First roll	Second roll	X	Y
1	1		
1	2		
1	3		
1	4		
2	1		
2	2		
2	3		
2	4		
3	1		
3	2		
3	3		
3	4		
4	1		
4	2		
4	3		
4	4		

Example 6.1. Roll a four-sided die twice, and record the result of each roll in sequence. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Table 6.1 represents the possible outcomes. Assume the die is fair and the rolls are independent.

1. Fill in the values of X and Y for each outcome in the table.

2. Identify the event $\{X = 4\}$ and compute its probability. Then interpret the probability both as a long relative frequency and as a relative likelihood.

3. Construct a table and plot of $\mathrm{P}(X = x)$ for each possible value x of X.

4. Construct a table and plot of $\mathrm{P}(Y = y)$ for each possible value y of Y.

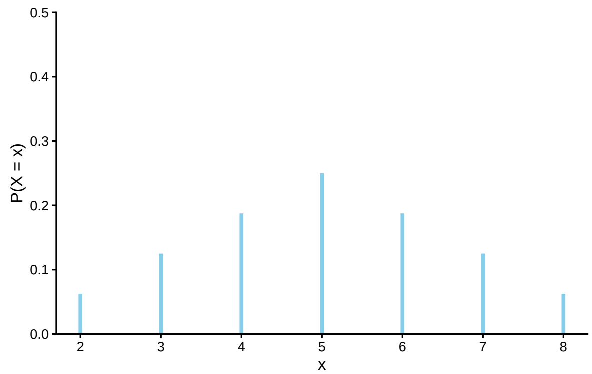


Figure 6.1: The marginal distribution of X, the sum of two rolls of a fair four-sided die.

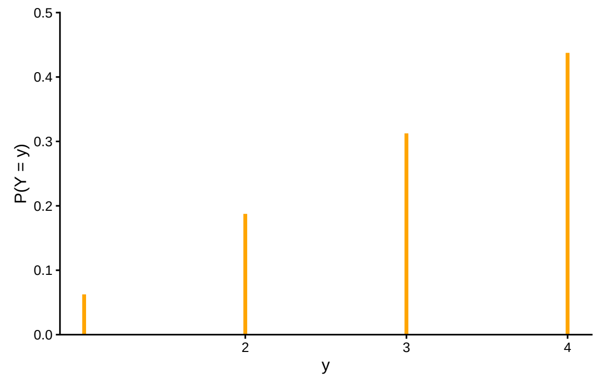


Figure 6.2: The marginal distribution of Y, the larger of two rolls of a fair four-sided die.

Example 6.2. Recall the batch testing scenarios from Example 5.2 and Example 5.3. Let X be the total number of tests required when we test all 8 people in a single batch first, and let Y be the total number of tests when we split into two batches of 4 people first.

1. Identify the distribution of X.

2. Identify the distribution of Y.

Example 6.3. The “matching problem” involves n distincts objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. Let X be the number of matches, that is, the number of objects for which their label matches the label of the box in which they are placed.

For example, suppose n people have a secret Santa gift exchange. Each person puts their name in a hat, the names are well shuffled, and everyone draws a name from the hat (to then purchase a gift for the selected person). Let X represent the number of people who draw their own name.

Consider the matching problem with $n = 3$. Specify the distribution of X, and interpret it.

6.1 Exercises

Exercise 6.1. Five individuals are interviewing for a single job opening, but only two are qualified for the position. Suppose the individuals are interviewed in a random order, and each individual is only interviewed once, until one of the qualified candidates is interviewed, and then no more candidates are interviewed. Let X count the number of interviews conducted.

Make a table to represent the distribution of X.

Exercise 6.2. Continuing Example 6.1.

1. Describe how you could simulate a single value of Y.

2. Construct a spinner (like from a kids game) to represent the distribution of Y.

3. Describe another way to simulate a single value of Y.

4. Describe how you could use simulation to approximate the distribution of Y.

- Don't confuse a random variable with its distribution!
  - A random variable measures a numerical quantity which depends on the outcome of a random phenomenon
  - The distribution of a random variable specifies the long run pattern of variation of values of the random variable over many repetitions of the underlying random phenomenon.
- Any marginal distribution can be represented by a single spinner (like from a kids game).
- In principle, there are always two ways of simulating a value x of a random variable X.
  1. *Simulate from the probability space.* Simulate an outcome ω from the underlying probability space and set $x = X(\omega)$.
  2. *Simulate from the distribution.* Construct a spinner corresponding to the distribution of X and spin it once to generate x.
- The second method requires that the distribution of X is known. However, as we will see in many examples, *it is common to specify the distribution of a random variable directly without defining the underlying probability space.*

7 Probability Mass Functions

- The **(probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).
- The probability distribution of a single discrete random variable X is often displayed in a table containing the probability of the event $\{X = x\}$ for each possible value x.
- In some cases, a distribution has a "formulaic" shape. For a discrete random variable X, the **probability mass function (pmf)** p_X expresses $\mathrm{P}(X = x)$ as a function of x: $p_X(x) = \mathrm{P}(X = x)$.
- Be sure to specify the possible values of the random variable!

Example 7.1. Continuing Example 6.1. Roll a four-sided die twice, and record the result of each roll in sequence. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Assume the die is fair and the rolls are independent.

Verify that the pmf of X is

$$p_X(x) = \frac{4 - |x - 5|}{16}, \qquad x = 2, 3, \ldots, 8$$

Example 7.2. Suppose a very large number of applicants have applied for a job, and you'll interview one applicant at a time until you find one that is qualified, whom you'll hire, and then you stop interviewing people. Let X be the total number of applicants interviewed (including the hire)

Assume that 30% of applicants are qualified, and applicants are interviewed one at a time in random order, independently.

1. Compute $\mathrm{P}(X = 1)$.

2. Compute $\mathrm{P}(X = 2)$.

3. Compute $\mathrm{P}(X = 3)$.

4. Compute $\mathrm{P}(X = 4)$.

5. Specify the probability mass function of X.

- In a sequence of **Bernoulli(p) trials**
  - There are only two possible outcomes, "success" (1) or not (0, "failure"), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to p.
  - The trials are independent.
- Perform Bernoulli(p) trials until a success occurs and then stop. Let X count the total number of trials, including the single success. The distribution of X is defined to be the **Geometric(p) distribution**.
- A discrete random variable X has a Geometric(p) distribution if and only if its probability

mass function is

$$p_X(x) = p(1-p)^{x-1}, \qquad\qquad x = 1, 2, 3, \dots$$

- The Geometric(p) probability mass function follows from the fact that if X has a Geometric distribution, then $X = x$ if and only if
  - the first $x-1$ trials are failures, and
  - the xth (last) trial results in success.

Example 7.3. Randomly select a county in the U.S. Let X be the leading digit in the county's population. For example, if the county's population is 10,040,000 (Los Angeles County) then $X = 1$; if 3,170,000 (Orange County) then $X = 3$; if 283,000 (SLO County) then $X = 2$; if 30,600 (Lassen County) then $X = 3$. The possible values of X are $1, 2, \dots, 9$. You might think that X is equally likely to be any of its possible values. However, a more appropriate model is to assume that X has pmf

$$p_X(x) = \begin{cases} \log_{10}(1 + \frac{1}{x}), & x = 1, 2, \dots, 9, \\ 0, & \text{otherwise} \end{cases}$$

This distribution is known as Benford's law.

1. Construct a table specifying the distribution of X, and the corresponding spinner.

2. Find $\mathrm{P}(X \geq 3)$

- Certain common distributions have special names and properties.
- Learn to identify these special situations and use their "shortcut" formulas.

7.1 Exercises

Exercise 7.1. Continuing Example 7.1. Is the following the pmf of Y? Discuss.

$$p_Y(u) = \frac{2u-1}{16}$$

Exercise 7.2. Shishito peppers are typically relatively mild, but about 10% are spicy. Suppose you eat Shishito peppers one at a time until you eat a spicy one and then you stop. Let X be the total number of peppers you eat (including the spicy one).

1. Explain why it is reasonable to assume eating the peppers constitutes a sequence of Bernoulli trials.

2. Describe in full detail how you could use simulation to approximate the distribution of X.

3. Identify the distribution of X by name, including the values of any relevant parameters.

4. Specify the probability mass function of X.

5. Make a table representing the distribution of X. (You can cut it off after the probabilities get small.)

6. Compute $\mathrm{P}(X > 10)$ without summing terms.

Exercise 7.3. Recall the matching problem from Example 6.3 with a general n and let X be the number of matches. Finding the distribution of X by enumerating the $n!$ possible outcomes as we did for $n = 3$ is not feasible in general. But we can approximate the distribution of X with simulation, as we did in an earlier exercise using this applet. Simulation shows that the distribution of the number of matches X is approximately the same for all n (unless n is very small, i.e., $n \leq 5$). In particular, for any $n > 5$, the probability mass function of X is approximately

$$p_X(x) = \frac{e^{-1}}{x!}, \qquad x = 0, 1, 2, ...$$

We will study where this distribution—called the Poisson(1) distribution—comes from in more detail later. For now we'll just use the pmf formula.

1. Construct a table, plot, and spinner corresponding to the above pmf. Compare to the simulation results for general n.

2. Use the pmf to approximate the probability of at least one match, and compare to the simulation results for general n.

3. Interpret the probability both as a long relative frequency and as a relative likelihood.

8 Expected Value

- The distribution of a random variable specifies the possible values and the probability of any event that involves the random variable.
- The distribution of a random variable contains all the information about its long run behavior. It is also useful to summarize some key features of a distribution.
- One summary characteristic of a distribution is the **long run average value** of the random variable.
- We can approximate the long run average value by simulating many values of the random variable and computing the average (mean) in the usual way: sum the simulated values and divide by the number of simulated values.

Example 8.1. This is a very simplified example illustrating the basic idea of how insurance works. Every year an insurance company sells many thousands of car insurance policies to drivers within a particular risk class. Each policyholder pays a "premium" of \$1000 at the start of the year, and the insurance company agrees to pay for the cost of all damages that occur during the year. Suppose that each policy incurs damage of either \$0, \$5000, \$20000, or \$50000 with the following probabilities.

Amount of damage (\$)	Profit (\$)	Probability
0	1000	0.910
5000	-4000	0.070
20000	-19000	0.019
50000	-49000	0.001

The insurance company's profit on a policy at the end of the year is the difference between the premium of \$1000 and any damage paid out. For example, a policy that incurs no damage results in a profit of \$1000; a policy that incurs \$5000 in damage results in a profit of -\$4000 (that is, a loss of \$4000) for the insurance company.

1. Interpret the probabilities 0.91, 0.07, 0.019, and 0.001 as long run relative frequencies in this context.

2. Imagine 100,000 hypothetical policies. How many of these policies would you expect to result in a profit of \$1000? -\$4000? -\$19000? -\$49000?

3. What do you expect the total profit for these 100,000 policies to be?

4. What do you expect the average profit per policy for these 100,000 policies to be?

5. Compute the probability that a policy has a profit equal \$220.

6. Compute the probability that a policy has a profit greater than \$220.

7. Is \$220 the most likely value of profit for a single policy?

8. Is \$220 the profit you would expect for a single policy?

9. Explain in what sense \$220 is "expected".

- The long run average value of a random quantity is called its "expected value".
- Be careful: the term "expected value" is somewhat of a misnomer.
- The expected value is *not* necessarily the value we expect on a single repetition of the random phenomenon, nor the most likely value (or even a possible value).
- Rather, the expected is the value we expect to see on average in the long run over many repetitions.
- A probability can be interpreted as a long run relative frequency; an expected value can be interpreted as a long run average value.

Example 8.2. Continuing Example 8.1. We considered what we would expect for 100000 hypothetical policies, but what about an unspecified large number of policies?

1. Imagine that we have recorded the profit for each of a large number of policies (not necessarily 100000). Explain in words the process by which you would compute the average profit per policy. (In other, more general, words: how do you compute an average of a list of numbers?)

2. Given that the profit of any policy is either 1000, -4000, -19000, or -49000, how could we simplify the calculation of the sum in the previous part? Write a general expression for the average profit per policy in this scenario.

3. What do you think the expression in the previous part converges to in the long run?

4. Explain how the value in the previous part is a "probability-weighted average value".

5. Compute the expected value of damage (not profit) as a probability-weighted average value.

6. Interpret the value from the previous part as a long run average value in this context.

7. How is the expected value of profit related to the expected value of damage? Does this make sense? Why?

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $\mathrm{E}(X)$ representing the probability-weighted average value of X.
- The expected value of a *discrete* random variable with pmf p_X is defined as

$$\mathrm{E}(X) = \sum_x x p_X(x) = \sum \text{value} \times \text{probability}$$

- Note well that $\mathrm{E}(X)$ represents *a single number*.
- The expected value is the "balance point" (center of gravity) of a distribution.
- The "probability-weighted average value" is just a more compact way of computing an average in the usual way: add up all the values and divide by the number of values.

- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $\mathrm{E}(\cdot)$ as
  - Simulate lots of values of what's inside $(\cdot)$
  - Compute the average. This is a "usual" average; just sum all the simulated values and divide by the number of simulated values.

Example 8.3. Recall Example 7.2. Suppose a very large number of applicants have applied for a job, and you'll interview one applicant at a time until you find one that is qualified, whom you'll hire, and then you stop interviewing people. Let X be the total number of applicants interviewed (including the hire)

Assume that 30% of applicants are qualified, and applicants are interviewed one at a time in random order, independently.

1. Make an educated guess for the expected value of X. Suggest an intuitive formula for the expected value of a Geometric(p) distribution. (It might help to consider $p = 0.1$ first.)

2. Use the definition to compute $\mathrm{E}(X)$. Does it match your guess?

3. Interpret $\mathrm{E}(X)$ in this context.

- If X has a Geometric distribution with parameter p then $\mathrm{E}(X) = 1/p$.

8.1 Exercises

Exercise 8.1. Continuing Example 8.1 and Example 8.2. For a randomly selected policy let X be the amount of damage and let Y be the company's profit.

1. What is the relationship between X and Y?

2. Compute and interpret $\mathrm{E}(Y)$.

3. What is the relationship between $\mathrm{E}(X)$ and $\mathrm{E}(Y)$? Does this make sense? Why?

- Expected values and linear rescaling: If X is a random variable and a, b are non-random constants then
$$\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$$

Exercise 8.2. Recall the matching problem from Example 6.3 and Exercise 7.3 with a general n and let X be the number of matches.

1. Compute and interpret $\mathrm{E}(X)$ when $n = 3$.

2. Compute and interpret $\mathrm{E}(X)$ for a general n.

9 Variance and Standard Deviation

- The expected value, a.k.a. long run average value, is just one feature of a distribution.
- Random variables vary, and the distribution describes the entire pattern of variability.
- Roughly, standard deviation measures overall degree of variability in a single number, as the average distance from the mean.
- Technically, the **variance** is the long run average squared distance from the mean, and the **standard deviation** is the square root of the variance.

Example 9.1. We'll motivate the calculation of variance by considering a small example. Say we have data on living area X, measure in thousands of square feet, for a small sample of 10 houses.

repetition	X	Deviation from mean	Squared deviation
1	1.12		
2	1.05		
3	1.67		
4	1.48		
5	0.43		
6	2.50		
7	1.71		
8	1.88		
9	0.85		
10	1.90		
Sum			
Average			

1. Compute the average of the X values.
2. Compute the deviation from the mean for each X value.
3. Compute the squared deviations.
4. Compute the average squared deviation. This is called the variance. What are the measurement units?
5. Compute the square root of the variance. This is called the standard deviation. What are the measurement units?

- The **variance** of a random variable X is

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{E}\left((X - \mathrm{E}(X))^2\right) \\ &= \mathrm{E}\left(X^2\right) - \left(\mathrm{E}(X)\right)^2\end{aligned}$$

- The **standard deviation** of a random variable is

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

- Variance is the long run average squared deviation from the mean.
- Standard deviation measures, roughly, the long run average distance from the mean. The measurement units of the standard deviation are the same as for the random variable itself.
- The definition $\mathrm{E}((X - \mathrm{E}(X))^2)$ represents the concept of variance. However, variance is usually computed using the following equivalent but slightly simpler formula.

$$\mathrm{Var}(X) = \mathrm{E}\left(X^2\right) - \left(\mathrm{E}\left(X\right)\right)^2$$

- That is, variance is the expected value of the square of X minus the square of the expected value of X.

- Variance has many nice theoretical properties. Whenever you need to compute a standard deviation, first find the variance and then take the square root at the end.

Example 9.2. An American roulette wheel has 18 black spaces, 18 red spaces, and 2 green spaces, all the same size and each with a different number on it. Xavier bets \$1 on black. If the wheel lands on black, Xavier wins his bet back plus an additional \$1; otherwise he loses the money he bet. Let X be Xavier's net winnings (net of the initial bet of \$1.)

1. Find the distribution of X.

2. Compute $\mathrm{E}(X)$.

3. Interpret $\mathrm{E}(X)$ in context.

4. An expected profit *for the casino* of 5 cents per \$1 bet seems small. Explain how casinos can turn such a small profit into billions of dollars.

5. Compute $\mathrm{Var}(X)$.

6. Compute and interpret $\mathrm{SD}(X)$.

Example 9.3. Continuing with roulette, Yasmin bets \$1 on number 7. If the wheel lands on 7, Yasmin wins her bet back plus an additional \$35; otherwise she loses the money she bet. Let Y be Yasmin's net winnings (net of the initial bet of \$1.)

1. Find the distribution of Y.

2. Compute $\mathrm{E}(Y)$.

3. How do the expected values of the two \$1 bets—bet on black versus bet on 7—compare? Does that mean that the two \$1 bets—bet on black versus bet on 7—are identical? If not, explain why not.

4. Before doing any calculations, determine if $\mathrm{SD}(Y)$ is greater than, less than, or equal to $\mathrm{SD}(X)$. Explain.

5. Compute $\mathrm{Var}(Y)$.

6. Compute and interpret $\mathrm{SD}(Y)$.

- Standardization measures values in terms of "standard deviations away from the mean"
- This idea is particularly useful when comparing random variables with different measurement units but whose distributions have similar shapes.

$$\text{Standardized value} = \frac{\text{Value - Mean}}{\text{Standard deviation}}$$

- If X is a random variable with expected value $\mathrm{E}(X)$ and standard deviation $\mathrm{SD}(X)$, then the **standardized random variable** is

$$Z = \frac{X - \mathrm{E}(X)}{\mathrm{SD}(X)}$$

Example 9.4. The SAT and ACT are standardized tests; scores range from 200 to 1600 for the SAT and from 1 to 36 for the ACT. SAT scores have, approximately, a symmetric bell-shaped distribution with a mean of 1050 and a standard deviation of 200. ACT scores have, approximately, a symmetric bell-shaped distribution with a mean of 21 and a standard deviation of 5.5. Troy's score on the SAT is 1500. Abed's score on the ACT is 31. Who scored relatively better on their test?

1. Compute and interpret the standardized value for Troy's SAT score.

2. Compute and interpret the standardized value for Abed's ACT score.

3. Who scored relatively better on their test?

9.1 Exercises

Example 9.5. The plots in Figure 9.1 summarize hypothetical distributions of quiz scores in six classes. All plots are on the same scale. Each quiz score is a whole number between 0 and 10 inclusive.

1. Donny Dont says that C represents the smallest SD, since there is no variability in the heights of the bars. Do you agree that C represents "no variability"? Explain.

2. What is the smallest *possible* value the SD of quiz scores could be? What would need to be true about the distribution for this to happen? (This scenario might not be represented by one of the plots.)

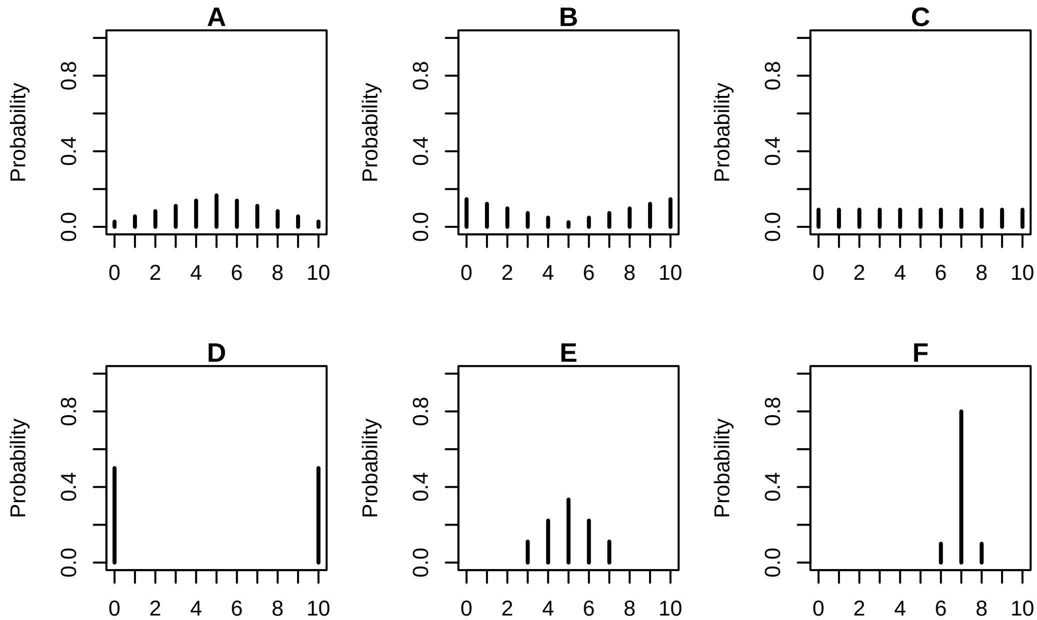


Figure 9.1: A few discrete distributions.

3. Without doing any calculations, arrange the classes in order based on their SDs from smallest to largest.

4. In one of the classes, the SD of quiz scores is 5. Which one? Why?

5. Is the SD in F greater than, less than, or equal to 1? Why?

6. Without doing any calculations, provide a ballpark estimate of SD in each case.

7. The standard deviation in plot E is about 1.15. Suppose that the scores are curved by adding 2 points to every score. What is the standard deviation and variance of the revised scores?

8. The standard deviation in plot E is about 1.15. Suppose that the scores are curved by multiplying every score by 2. What is the standard deviation and variance of the revised scores?

- Variance and linear rescaling: If X is a random variable and a, b are non-random constants then

$$\begin{aligned}
\mathrm{SD}(aX + b) &= |a|\mathrm{SD}(X) \\
\mathrm{Var}(aX + b) &= a^2\mathrm{Var}(X)
\end{aligned}$$

Exercise 9.1. Recall the matching problem from Example 6.3, Exercise 7.3, and Exercise 8.2 with a general n and let X be the number of matches.

1. Compute and interpret $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$ when $n = 3$.

2. Compute and interpret $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$ for a general n.

Exercise 9.2. Suppose that X has a Geometric(p) distribution.

1. Without doing any computations, determine $\mathrm{Var}(X)$ if $p = 1$.

2. As p decreases, does $\mathrm{Var}(X)$ increase or decrease? Why?

- If X has a Geometric distribution with parameter p then $\mathrm{Var}(X) = (1 - p)/p^2$.

10 Binomial Distributions

- Some probabilistic situations are so common that they have special names.
- In a sequence of **Bernoulli(p) trials**
  - There are only two possible outcomes, "success" (1) or not (0, "failure"), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to p.
  - The trials are independent.
- There are several commonly used distributions associated with Bernoulli trials, including Geometric and Binomial.

Example 10.1. Consider an extremely simplified model for the daily closing price of a certain stock. Every day the price either goes up or goes down, and the movements are independent from day-to-day. Assume that the probability that the stock price goes up on any single day is 0.55. Let X be the number of days on which the price goes up in a 5-day week.

1. Compute and interpret $\mathrm{P}(X = 0)$.

2. Compute the probability that the price goes up on the first day and then down on the following four days.

3. Why is $\mathrm{P}(X = 1)$ different from the probability in the previous part? Compute and interpret $\mathrm{P}(X = 1)$.

4. Compute and interpret $\mathrm{P}(X = 2)$.

5. Suggest a general formula for the probability mass function of X.

6. Construct a table and plot to represent the distribution of X, and interpret it.

- Consider a *fixed number*, n, of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
- A Binomial distribution is specified by two parameters:
  - n (an integer): the fixed number of Bernoulli trials, or the sample size
  - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Binomial distribution** with parameters n, a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad\qquad x = 0, 1, 2, \dots, n$$

  - In the Binomial(n, p) situation, there are $\binom{n}{x}$ S/F sequences that result in exactly x successes—and thus $n - x$ failures—in n trials,
  - and each of these sequences has probability $p^x(1-p)^{n-x}$
- The *binomial coefficient* (read "n choose x")

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

counts the number of success/failure sequences of length n in which there are exactly x successes. (Remember: by definition $0! = 1$.)

Example 10.2. Continuing Example 10.1.

1. Suggest a shortcut formula for $\mathrm{E}(X)$.

2. Compute $\mathrm{E}(X)$ using the distribution of X from Example 10.1. Did the shortcut formula work?

3. Interpret $\mathrm{E}(X)$.

4. Remember that we assumed $p = 0.55$. If instead p were 0.9, would $\mathrm{Var}(X)$ be greater or less? What if $p = 0.5$?

- If X has a Binomial(n, p) distribution then

$$\begin{aligned}\mathrm{E}(X) &= np \\ \mathrm{Var}(X) &= np(1-p)\end{aligned}$$

10.1 Exercises

Exercise 10.1. Continuing Example 10.1.

1. What does the random variable $5 - X$ represent? What is its distribution?

2. Suppose that the price is currently \$100 and each it either moves up \$2 or down \$2. Let S be the stock price after 5 days. How does S relate to X? Does S have a Binomial distribution?

3. Recall that X is the number of days on which the price goes up in the next five days. Suppose that Y is the number of days on which the price goes up in the ten days after that (days 6-15). What is the distribution of Y? What does $X + Y$ represent and what is its distribution? (Continue to assume independence between days, with probability 0.55 of an up movement on any day.)

- If X and Y are independent, X has a Binomial(n_X, p) distribution, and Y has a Binomial(n_Y, p) distribution then $X + Y$ has a Binomial($n_X + n_Y$, p)
- A Binomial(1, p) distribution is also known as a **Bernoulli(p)** distribution, taking a value of 1 with probability p and 0 with probability $1 - p$.
  - For a single trial, a Bernoulli random variable *indicates* if there is a success (1) or failure (0) on the trial
- If $X_1, X_2, \ldots, X_n$ are independent each with a Bernoulli(p) distribution, then $X_1 + \cdots + X_n$ has a Binomial(n, p) distribution.
  - The random variable $X_1 + X_2 + \ldots + X_n$ incrementally counts the total number of successes in the n trials

Exercise 10.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p. If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.

2. Roll a die 20 times; X is the number of times the die lands on 6.

3. Roll a die until it lands on 6; X is the total number of rolls.

4. Roll a die 20 times; X is the sum of the numbers rolled.

5. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)

6. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).

7. Randomly select a sample of 35 Cal Poly students; X is the number of students in the sample who are CA residents.

- Consider selecting a random sample of size n from a population in which every individual is classified as “success” or “failure”. Suppose that the proportion of individuals in the population classified as success is p.
- The individuals selected for the sample can be considered a sequence of Bernoulli(p) trials.
  - There are only two possible outcomes, success (1) and failure (0), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to the population proportion of success p
  - The trials are independent
    * If sampling with replacement, or
    * If sampling without replacement but the population size is much larger than the sample size n (which is usually the case in practice)
- If X counts the number of successes in a random sample of size n from a large population with population proportion of success p then X has a Binomial(n, p) distribution.

Exercise 10.3. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution*. Can you think of X and Y that have the same distribution but for which $\mathrm{P}(X = Y) = 0$?

11 Poisson Distributions

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location.

Example 11.1. Recall the matching problem from Example 6.3, Exercise 7.3, Exercise 8.2, and Exercise 9.1. There are n distinct objects that are shuffled and placed uniformly at random in n distinct spots with one object per spot, and X is the number of correct matches of object with spot. Note: n is a fixed and known number, but we are considering different values of it.

We have seen that $\mathrm{E}(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$), $\mathrm{Var}(X) = 1$ and the approximate distribution of X is given by Table 11.1.

Starting with $x = 0$, describe the distribution of X in terms of relative likelihoods:

1. 1 is [blank] times as likely as 0

2. 2 is [blank] times as likely as 1

3. 3 is [blank] times as likely as 2

4. 4 is [blank] times as likely as 3

Table 11.1: Poisson(1) distribution, the approximate distribution of X, the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.36788
1	0.36788
2	0.18394
3	0.06131
4	0.01533
5	0.00307
6	0.00051
7	0.00007

5. 5 is [blank] times as likely as 4

6. 6 is [blank] times as likely as 5

7. In general, x is [blank] times as likely as $x-1$, for $x = 1, 2, 3, ...$

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, ...$ and the pmf follows the

$$\text{Poisson}(\mu) \text{ pattern:} \quad x \text{ is } \frac{\mu}{x} \text{ as likely as } x-1 \qquad (\text{for } x = 1, 2, 3, ...)$$

- As a function of $x = 0, 1, 2, ...$, a Poisson(μ) pmf increases for $x < \mu$ and then decreases for $x > \mu$
  - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
  - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.

- If X has a Poisson(μ) distribution then

$$\begin{aligned}
\mathrm{E}(X) &= \mu \\
\mathrm{Var}(X) &= \mu
\end{aligned}$$

Example 11.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Technically, there is no fixed upper bound on what X can be, so mathematically it is convenient to consider $0, 1, 2, \ldots$ as the possible values of X. Assume that X has a Poisson distribution with parameter 2.4.

1. Interpret 2.4 in context.

2. Identify what the Poisson(2.4) patterns implies in context.

3. Sketch a plot of the distribution of X.

4. Assuming that the probability that X is greater than 9 is negligible, make a table representing the distribution of X.

5. The pmf of X is

$$\mathrm{P}(X = x) = e^{-2.4}\frac{2.4^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Use the pmf to compute $\mathrm{P}(X = 0)$.

6. Use the pmf to compute $\mathrm{P}(X = 1)$.

7. Use the pmf to compute $\mathrm{P}(X = 2)$.

8. Use the pmf to construct a table and plot representing the distribution of X. Does it match your previous work?

9. Use the pmf to compute $\mathrm{E}(X)$ and verify that it is 2.4.

10. Compute and interpret $\mathrm{P}(X \leq 2)$.

- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, ...$$

- The Poisson pattern implies that the shape of a Poisson pmf as a function of x is given by $\mu^x/x!$.

$$p(x) = p(0)\frac{\mu^x}{x!}, \qquad x = 0, 1, 2, ...$$

- The constant $p(0) = e^{-\mu}$ simply renormalizes the pmf so that the probabilities sum[1] to 1.

Example 11.3. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. In what ways is this like the Binomial situation?

2. In what ways is this NOT like the Binomial situation?

3. In what sense are home runs "relatively rare"?

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location
- **Poisson paradigm.** Let $A_1, A_2, \dots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = \mathrm{P}(A_i)$. Let $N = \mathrm{I}_{A_i} + \mathrm{I}_{A_2} + \cdots + \mathrm{I}_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose
  - n is "large" (but not necessarily fixed),

[1]Recall the series expansion $e^\mu = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

  - $p_1, \dots, p_n$ are "comparably small" (but not necessarily the same), and
  - the events $A_1, \dots, A_n$ are "not too dependent" (but not necessarily independent).
- Then N has an approximate Poisson distribution with parameter $\mathrm{E}(N) = \sum_{i=1}^{n} p_i$.

11.1 Exercises

Exercise 11.1. Continuing Example 11.1. Explain why X in the matching problem has a Poisson distribution when n is large.

Exercise 11.2. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a Poisson(0.53) distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a "month").

1. Make an educated guess for $\mathrm{E}(X)$.

2. Compute $\mathrm{P}(X = 0)$.

3. Compute $\mathrm{P}(X = 1)$.

4. How could you use simulation to approximate the distribution of X?

5. Use simulation to approximate the distribution of X. Does the approximate distribution follow (approximately) the Poisson pattern?

6. Use a Poisson pmf to compute $\mathrm{P}(X = 5)$.

- **Poisson aggregation.** If X and Y are independent, X has a Poisson(μ_X) distribution, and Y has a Poisson(μ_Y) distribution, then $X + Y$ has a Poisson($\mu_X + \mu_Y$) distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.

12 Continuous Random Variables

- A continuous random variable can take any value within some uncountable interval.
- Simulated values of a continuous random variable are usually plotted in a **histogram** which groups the observed values into "bins" and plots densities or frequencies for each bin.
- In a histogram *areas* of bars represent relative frequencies; the axis which represents the height of the bars is called "density".
- The continuous analog of a probability mass function (pmf) for a discrete random variable is a probability density function (pdf) However there are some fundamental differences between discrete and continuous random variables, and pmfs and pdfs.
- The distribution of a continuous random variable can be described by a **probability density function (pdf)**, for which *areas under the density curve determine probabilities.*

Example 12.1. Suppose that in your office everyone arrives to work between 8am and 9am. Randomly select an employee and let X be employee's arrival time measured in minutes after 8am, including fractions of a minute, so that X takes a value in the continuous interval [0, 60].

Data on many randomly selected employees suggests that the distribution of X is described by this probability density function (pdf), where c is a constant that we'll determine soon.

$$f_X(x) = \begin{cases} cx, & 0 < x < 60 \\ 0, & \text{otherwise.} \end{cases}$$

1. Sketch a plot of the pdf. Roughly, what does this say about arival times?

2. Find the value of c and specify the pdf of X.

3. Compute and interpret $\mathrm{P}(X < 15)$.

4. Compute and interpret $\mathrm{P}(X > 45)$.

- The continuous analog of a probability mass function (pmf) is a *probability density function (pdf)*.
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- A pdf of a continuous random variable must be *integrated* to find probabilities of related events.
- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$\mathrm{P}(a \le X \le b) = \int_a^b f_X(x)dx, \qquad \text{for all } -\infty \le a \le b \le \infty \tag{12.1}$$

- For a continuous random variable X with pdf f_X, the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \ge 0 \qquad \text{for all } x, \tag{12.2}$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{12.3}$$

Example 12.2. Continuing Example 12.1, recall that X has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 \le x \le 60, \\ 0, & \text{otherwise.} \end{cases}$$

1. Consider the probability that an employee arrives within 1 minute after 8:15. Compute $\mathrm{P}(15 < X < 16)$, the probability that X, truncated to the nearest minute, is 15. How is the probability related to $f(15)$?

2. Consider the probability that an employee arrives within 1 minute after 8:45. Compute $\mathrm{P}(45 < X < 46)$, the probability that X, truncated to the nearest minute, is 45. How is the probability related to $f(45)$?

3. Compute and interpret the ratio of the two previous probabilities.

4. Consider the probability that an employee arrives within 1 *second* after 8:15. Compute $\mathrm{P}(15 < X < 15 + 1/60)$, the probability that X, truncated to the nearest *second*, is 15. (The measurement units of X are minutes, so $1/60$ minutes is 1 second.) How is the probability related to $f(15)$?

5. Consider the probability that an employee arrives within 1 *second* after 8:45. Compute $\mathrm{P}(45 < X < 45 + 1/60)$, the probability that X, truncated to the nearest *second*, is 45. How is the probability related to $f(45)$?

6. Compute and interpret the ratio of the two previous probabilities.

7. Consider the probability that an employee arrives within 1 *millisecond* after 8:15. Compute $\mathrm{P}(15 < X < 15 + 1/60000)$, the probability that X, truncated to the nearest *millisecond*, is 15. (The measurement units of X are minutes, so $1/60000$ minutes is 1 millisecond.) How is the probability related to $f(15)$?

8. Consider the probability that an employee arrives within 1 *millisecond* after 8:45. Compute $\mathrm{P}(45 < X < 45 + 1/60000)$, the probability that X, truncated to the nearest *millisecond*, is 45. How is the probability related to $f(45)$?

9. Compute and interpret the ratio of the two previous probabilities.

10. Compute and interpret $\mathrm{P}(X = 15)$.

11. Compute and interpret $\mathrm{P}(X = 45)$.

12. In practical terms, is any employee more likely to arrive "at 8:15" or "at 8:45"? Explain.

- **The probability that a continuous random variable X equals any particular value is 0**. That is, if X is continuous then $\mathrm{P}(X = x) = 0$ for all x.
- For a *continuous* random variable X, $\mathrm{P}(X \le x) = \mathrm{P}(X < x)$.
  - Be careful: for a *discrete* random variable X, $\mathrm{P}(X \le x)$ can be different than $\mathrm{P}(X < x)$.
- Even though any specific value of a continuous random variable has probability 0, *intervals* still can have positive probability.
- For continuous random variables, it doesn't really make sense to talk about the probability that the random value is *equal to* a particular value. In practical applications involving continuous random variables, "equal to" really means "close to", and "close to" probabilities correspond to intervals which can have positive probability.

- The density $f_X(x)$ at value x is *not* a probability.
- Rather, the density $f_X(x)$ at value x is related to the probability that the RV X takes a value "close to x" in the following sense

$$\mathrm{P}\left(x - \frac{\epsilon}{2} \le X \le x + \frac{\epsilon}{2}\right) \approx f_X(x)\epsilon, \qquad \text{for small } \epsilon$$

- The quantity ϵ is a small number that represents the desired degree of precision. For example, rounding to two decimal places corresponds to $\epsilon = 0.01$.
- Roughly,

$$\mathrm{P}(X \text{ is close to } x) \approx f_X(x) \times (\text{what counts as close to})$$

- What's important about a pdf is *relative* heights. For example, if $f_X(x_2) = 2f_X(x_1)$ then X is roughly "twice as likely to be close to x_2 than to be close to x_1" in the above sense.

$$\frac{f_X(x_2)}{f_X(x_1)} = \frac{f_X(x_2)\epsilon}{f_X(x_1)\epsilon} \approx \frac{\mathrm{P}\left(x_2 - \frac{\epsilon}{2} \le X \le x_2 + \frac{\epsilon}{2}\right)}{\mathrm{P}\left(x_1 - \frac{\epsilon}{2} \le X \le x_1 + \frac{\epsilon}{2}\right)} \approx \frac{\mathrm{P}(X \text{ is close to } x_2)}{\mathrm{P}(X \text{ is close to } x_1)}$$

Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Sketch the pdf of X and interpret, roughly, its shape in context.

2. Without integrating, approximate the probability that X rounded to the nearest hundred dollars is \$500.

3. Compute and interpret the ratio $f(0.5)/f(5)$.

4. Compute and interpret $\mathrm{P}(X < 5)$.

5. Compute and interpret $\mathrm{P}(X > 10)$.

12.1 Exercises

Exercise 12.1. In a certain population, household income X ($ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are $ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. What are the possible values of X? Sketch the pdf of X.

2. Without integrating, find the probability that a household has an income, rounded to the nearest thousand dollars, of $200K.

3. Without integrating, find the probability that a household has an income, rounded to the nearest *hundred* dollars, of $200K.

4. Without integrating, determine how many times more likely it is for a household to have an income, rounded to the nearest *hundred* dollars, of $100K than $200K.

5. Compute the value of c. (For this and the following parts, set up any calculations you would need to perform, but you can use a calculator or software to do the computations.)

6. Compute the probability that a household has an income under \$100K.

7. Compute the probability that a household has an income over \$200K.

8. Find the probability that a household has an income exactly equal to \$200K.

Exercise 12.2. In Example 12.1, assume instead that employees are "equally likely" to arrive at any time between 8am and 9am. Let X be the arrival time of a randomly selected employee, measured in minutes after 8am. We'll assume that X has a "Uniform distribution" on [0, 60].

1. What shape would you want the pdf to have? Sketch that shape and then determine the pdf.

2. Compute and interpret $\mathrm{P}(X < 15)$.

3. Compute and interpret $\mathrm{P}(X > 45)$.

4. How do the two previous parts compare to each other? Why? How do they compare to the corresponding parts in Example 12.1? Why?

5. Compute $\mathrm{P}(X = 15)$.

6. Replace the previous part with a more sensible question, then solve it.

7. Make an educated guess for $\mathrm{E}(X)$.

8. Make an educated guess for $\mathrm{SD}(X)$. (We'll see how to compute expected value and variance for continuous random variables later, but for this example you should be able to make reasonable guesses without doing much computation.)

- A continuous random variable X has a **Uniform distribution** with parameters a and b, with $a < b$, if its probability density function f_X satisfies

$$
\begin{aligned}
f_X(x) &\propto \text{constant}, && a < x < b && (12.4)\\
&= \frac{1}{b-a}, && a < x < b. && (12.5)
\end{aligned}
$$

- If X has a Uniform(a, b) distribution then

$$
\begin{aligned}
\mathrm{E}(X) &= \frac{a+b}{2} && (12.6)\\
\mathrm{Var}(X) &= \frac{|b-a|^2}{12} && (12.7)\\
\mathrm{SD}(X) &= \frac{|b-a|}{\sqrt{12}} \approx 0.289|b-a| && (12.8)
\end{aligned}
$$

13 Percentiles: Cumulative Distribution Functions and Quantile Functions

- The probability density function of a continuous random variable describes which values are more or less likely in the "close to" sense.
- However, the pdf does not provide probabilities directly; rather, it is areas under the pdf which provide probabilities.
- There are many ways of describing distributions. In particular, a distribution is basically just a collection of percentiles.
- Cumulative distribution functions and quantile functions describe distributions by specifying all the percentiles.
- Roughly,
  - The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile.
  - The *quantile function* fills in the following blank for a given $p \in (0, 1)$: the $(100p)$th percentile is (blank).

Example 13.1. Explain what the percentiles mean in the following statements.

- 47cm is 12th percentile of heights for newborn girls.
- 620 is the 80th percentile of SAT Math scores.

- A distribution is characterized by its percentiles.
- Roughly, the value x is the p-th **percentile** (a.k.a. quantile) of a distribution of a random variable X if p percent of values of the variable are less than or equal to x: $\mathrm{P}(X \le x) = p$.
- A few commonly used percentiles
  - First (or lower) quartile is the 25th percentile ($p = 0.25$)
  - Median is the 50th percentile ($p = 0.50$)

- Third (or upper) quartile is the 75th percentile ($p = 0.75$)

Example 13.2. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute $\mathrm{P}(X < 15)$ and phrase the result in terms of a percentile.

2. Compute $\mathrm{P}(X < 45)$ and phrase the result in terms of a percentile.

3. Find an expression for $\mathrm{P}(X \leq x)$ for general x. This function, denoted $F_X(x)$, is called the *cumulative distribution function (cdf)* of X.

4. How does the cdf relate to percentiles?

5. Use the cdf to determine what percentile 30 minutes after 8am is.

6. Compute and interpret $F(40) - F(30)$.

- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile. That is, for an input x, the cdf outputs $\mathrm{P}(X \le x)$.
- The **cumulative distribution function (cdf)** of a random variable X is the function F_X defined by $F_X(x) = \mathrm{P}(X \le x)$. A cdf is defined for all real numbers x regardless of whether x is a possible value of X.

Example 13.3. Continuing Example 13.2 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \qquad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

1. Find and interpret the 10th percentile.

2. Find and interpret the 90th percentile.

3. Find an equation for the $(100p)$th percentile (e.g., $p = 0.9$ for the 90th percentile).

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in (0, 1)$: the $100p$th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- For a *continuous* random variable X with cdf F_X, the **quantile function** Q_X is the inverse of the cdf, $Q_X(p) = F_X^{-1}(p)$.
- A quantile function basically gives you the same information as a cdf, just in the other direction.

- Both a quantile function and a cdf define a distribution by specifying all the percentiles.

Example 13.4. Continuing Example 13.3 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \qquad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

and quantile function

$$Q(p) = 60\sqrt{p}, \qquad 0 < p < 1.$$

1. Construct a circular spinner for simulating values of X. Version 1: label the values 0, 10, 20, 30, 40, 50, 60. Careful: they will not be equally spaced.

2. Sketch a histogram with equal bin widths—0 to 10, 10 to 20, etc.—that you would expect to see after spinning the version 1 spinner many times. Does it have the proper shape?

3. Construct a circular spinner for simulating values of X. Version 2: label the circular axis so that your spinner has 12 sectors each with probability 1/12.

4. Sketch a histogram that you would expect to see after spinning the version 2 spinner many times. Start by sketching a histogram with *unequal* bin widths, then sketch the correspond equal bin width histogram. Does it have the proper shape?

- The cdf and the quantile function can be used to create a spinner for a distribution.
- A spinner basically describes a distribution by specifying all the percentiles. For example,
  - the 25th percentile goes 25% of the way around ("3 o clock")
  - the 50th percentile goes 50% of the way around ("6 o clock")
  - the 75th percentile goes 75% of the way around ("9 o clock")
- Intervals corresponding to regions of higher density ("more likely" values) are stretched out on the spinner boundary; intervals corresponding regions of lower density ("less likely" values) are shrunk.
- **Universality of the Uniform (or "one spinner to rule them all").** Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F.
- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.
- Actually, universality of the uniform says we don't have to create a new spinner. We can just spin the Uniform(0, 1) spinner, with evenly spaced values from 0 to 1, and transform each resulting value by plugging it into the quantile function.

Example 13.5. Continuing Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Compute $\mathrm{P}(X < 5)$ and phrase the result in terms of a percentile.

2. Find an expression for the CDF of X.

3. Compute the 25th percentile.

4. Find an expression for the quantile function of X.

5. Use the quantile function to find the 75th percentile.

13.1 Exercises

Exercise 13.1. Continuing Exercise 12.1. In a certain population, household income X ($ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are $ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $\mathrm{P}(X \leq 100)$ and phrase the result in terms of a percentile.

2. Find the cdf of X.

3. Find the 10th percentile.

4. Find the quantile function of X.

5. Sketch a circular spinner for simulating values of X.

Exercise 13.2. Let X be the number of heads in 3 flips of a fair coin.

1. Find the pmf of X and sketch a plot of it.

2. Find the cdf of X and sketch a plot of it.

3. Let Y be the number of tails in 3 flips of a fair coin. Find the cdf of Y.

- A cdf is a non-decreasing function: if $x_1 \le x_2$ then $F_X(x_1) \le F_X(x_2)$.
- A cdf approaches 0 as the input approaches $-\infty$: $\lim_{x\to-\infty} F_X(x) = 0$
- A cdf approaches 1 as the input approaches ∞: $\lim_{x\to\infty} F_X(x) = 1$
- The cdf of a *discrete* random variable is a step function.
  - The steps occur at the possible values of the random variable.
  - The height of a particular step corresponds to the probability of that value, given by the pmf.
- The cdf of a *continuous* random variable is a continuous function.
  - The cdf of a *continuous* random variable is obtained by integrating the pdf, so

  - The pdf of a *continuous* random variable is obtained by differentiating the cdf

$$F'_X = f_X \qquad \text{if } X \text{ is continuous}$$

- For any random variable X with cdf F_X

$$F_X(b) - F_X(a) = \mathrm{P}(a < X \leq b)$$

  Whether the inequalities in the above event are strict ($<$) or not ($\leq$) matters for discrete random variables, but not for continuous.
- Random variables X and Y **have the same distribution** if their cdfs are the same, that is, if $F_X(u) = F_Y(u)$ for all u.
- That is, two random variables have the same distribution if all the percentiles are the same.

14 Expected Values of Continuous Random Variables

Example 14.1. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Explain how you could approximate $\mathrm{E}(X)$ as a sum (as if X were a discrete random variable). Hint: recall Example 12.2.

3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

4. Compute $\mathrm{E}(X)$ as an integral. Compare to the approximations.

5. Interpret $\mathrm{E}(X) = 40$ in context.

6. Compute and interpret $\mathrm{P}(X \le \mathrm{E}(X))$.

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $\mathrm{E}(X)$ representing the probability-weighted average value of X.
- The expected value of a *continuous* random variable with pdf f_X is defined as

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $\mathrm{E}(X)$ represents *a single number*.
- The expected value is the "balance point" (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $\mathrm{E}(\cdot)$ as
  - Simulate lots of values of what's inside $(\cdot)$
  - Compute the average. This is a "usual" average, regardless of whether the variable is discrete or continuous; just sum all the simulated values and divide by the number of simulated values.

Example 14.2. Continuing Example 14.1 where X has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

Now we want to compute variance and standard deviation.

1. What is the formula for computing variance?

2. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

3. Explain how you could approximate $\mathrm{E}(X^2)$ as a sum (like a discrete random variable).

4. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

5. Compute $\mathrm{E}(X^2)$ as an integral. (You should get 1800.)

6. Compute $\mathrm{Var}(X)$.

7. Compute and interpret $\mathrm{SD}(X)$.

- The **"law of the unconscious statistician" (LOTUS)** says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \qquad \mathrm{E}[g(X)] = \sum_x g(x)p_X(x)$$

$$\text{Continuous } X \text{ with pdf } f_X: \qquad \mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

- LOTUS says we don't have to first find the distribution of $Y = g(X)$ to find $\mathrm{E}[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X)$$

- In terms of expected values, in general

$$\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$$

The left side $\mathrm{E}(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(\mathrm{E}(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 14.3. Continuing Example 12.3. Suppose that X is a claim amount in excess of the decuctible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \qquad x > 0$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Donny Dont says $\mathrm{E}(X) = \int_0^\infty (1/4.3)e^{-x/4.3}dx = 1$. Do you agree?

3. Compute and interpret $\mathrm{E}(X)$.

4. Compute and interpret $\mathrm{P}(X \le \mathrm{E}(X))$.

5. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

6. Donny Dont says: "I can just use LOTUS and replace x with x^2, so $\mathrm{E}(X^2)$ is $\int_{-\infty}^{\infty}(1/4.3)x^2 e^{-x^2/4.3}dx$". Do you agree?

7. Compute $\mathrm{E}(X^2)$.

8. Compute $\mathrm{Var}(X)$.

9. Compute and interpret $\mathrm{SD}(X)$.

14.1 Exercises

Exercise 14.1. Continuous Exercise 12.1. In a certain population, household income X ($ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c. Note that the measurement units are $ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $\mathrm{E}(X)$.

2. Interpret $\mathrm{E}(X)$ in context.

3. Compute and interpret $\mathrm{P}(X \leq \mathrm{E}(X))$.

4. Compute $\mathrm{Var}(X)$.

5. Compute and interpret $\mathrm{SD}(X)$.

15 Exponential Distributions

- Exponential distributions are often used to model the waiting times between events in a random process that occur continuously over time.

Example 15.1. A business is tracking when online orders are placed on their website. Suppose an order just occurred and let X be the time until the next order is placed, measured in hours, including fractions of an hour. Assume that X has an *Exponential distribution with rate parameter 2.5 per hour*, with pdf

$$f_X(x) = 2.5e^{-2.5x}, \qquad x > 0$$

1. Sketch the pdf of X. Roughly, what does this tell you about times between orders?

2. Compute and interpret $\mathrm{P}(X < 0.5)$.

3. Determine the cdf of X.

4. Determine what percentile 2 hours is.

5. Compute and interpret the median of X.

6. Make an educated guess for $\text{E}(X)$. Then compute and interpret $\text{E}(X)$; did your guess work?

7. Compute and interpret $\text{P}(X < \text{E}(X))$.

- A continuous random variable X has an **Exponential distribution** with *rate* parameter $\lambda > 0$ if its pdf is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & \text{otherwise} \end{cases}$$

- If X has an Exponential(λ) distribution then

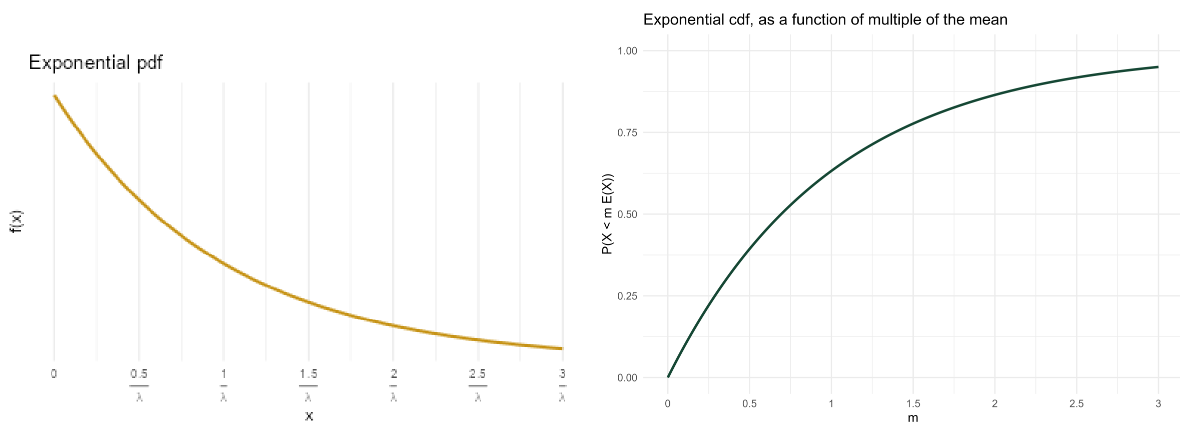
$$\begin{aligned} \text{cdf:} \quad F_X(x) &= 1 - e^{-\lambda x}, \quad x > 0 \\ \text{P}(X > x) &= e^{-\lambda x}, \quad x > 0 \\ \text{quantile:} \quad Q(p) &= -(1/\lambda)\ln(1-p), \quad 0 < p < 1 \\ \text{E}(X) &= \frac{1}{\lambda} \\ \text{Var}(X) &= \frac{1}{\lambda^2} \\ \text{SD}(X) &= \frac{1}{\lambda} \end{aligned}$$

- Percentiles for any Exponential distribution follow a particular pattern in terms of "multiples of the mean":
  - For any $m > 0$, $\text{P}(X \le m\text{E}(X)) = 1 - e^{-m}$.
    * For example, for any Exponential distribution the mean ($m = 1$) is the 63.2rd percentile ($1 - e^{-1} = 0.632$).
  - The pth percentile is $-\ln(1-p)$ times $\text{E}(X)$.

* For example, for any Exponential distribution the median ($p = 0.5$) is 0.693 times the mean ($-\ln(1-0.5) = 0.693$).

- See Table 15.1.

- Exponential distributions are often used to model the *waiting time* in a random process until some event occurs.
  - λ is the average *rate* at which events occur over time (e.g., 2 per hour)
  - $1/\lambda$ is the mean time between events (e.g., 1/2 hour)
- If X has an Exponential(λ) distribution and $a > 0$ is a constant, then aX has an Exponential(λ/a) distribution.
  - If X is measured in hours with rate $\lambda = 2$ per hour and mean 1/2 hour
  - Then $60X$ is measured in minutes with rate 2/60 per minute and mean $60(1/2) = 30$ minutes.
- If X has an Exponential(1) distribution and $\lambda > 0$ is a constant then X/λ has an Exponential(λ) distribution.
- If U has a Uniform(0, 1) distribution then $-\ln(1-U)$ has Exponential(1) distribution, and $-(1/\lambda)\ln(1-U)$ has an Exponential(λ) distribution.



(a) PDF (b) CDF

Figure 15.1: A generic Exponential distribution with rate parameter λ and mean $1/\lambda$. In the PDF plot, the x-axis is labeled in measurement units; in the CDF plot, the x-axis is labeled in terms of multiples of the mean.

Example 15.2. Continuing Example 15.1. Let $N(1)$ be the number of orders in a 1 hour period, a discrete random variable. Then the expected value of $N(1)$ is 2.5 orders. If $N(2)$ is the number of orders that happen in a 2 hour period then its expected value is $\mathrm{E}(N(2)) = 5$

Table 15.1: Percentiles for an Exponential distribution in terms of multiples of the mean

Percentile	Relative to mean
10%	0.11 times the mean
25%	0.29 times the mean
39.3%	0.50 times the mean
50%	0.69 times the mean
63.2%	1.00 times the mean
75%	1.39 times the mean
77.7%	1.50 times the mean
86.5%	2.00 times the mean
90%	2.30 times the mean
91.8%	2.50 times the mean
95%	3.00 times the mean
97%	3.50 times the mean
98.2%	4.00 times the mean
98.9%	4.50 times the mean
99.3%	5.00 times the mean

orders. In general, if $N(t)$ is the number of orders in a t hour period its expected value is $\mathrm{E}(N(t)) = 2.5t$ orders.

Assume that, for any t, the discrete random variable $N(t)$ has a Poisson distribution with parameter $2.5t$. Now we'll consider what this assumption implies about the time between orders. Suppose an order just occurred and let T be the time in hours until the next order; T is a continuous random variable.

1. Interpret the event $\{T > 0.5\}$. How can you express this as an equivalent event involving a number of order?

2. Compute and interpret $\mathrm{P}(T > 0.5)$.

3. Compute $\mathrm{P}(T > t)$ as a function of $t > 0$.

4. Identify by name the distribution of T; be sure to specify the values of any relevant parameters.

5. Compute and interpret $\mathrm{E}(T)$.

- Events occur continuously over time according to a **Poisson process** with *rate* $\lambda > 0$ (per unit time) if
  - The number of events that occur in any time interval of length t has a Poisson(λt) distribution.
    - In particular, the distribution does not depend on the starting time or ending time of the interval, just its length.
  - The numbers of events that occur in disjoint intervals of time are independent.
- A random process is a Poisson process with rate λ if and only if
  - it is a counting process, that is, it is a process which counts the number of occurrences of some event over time.
  - the *interarrival times* (times between events) $W_0, W_1, \ldots$ are independent Exponential(λ) random variables

15.1 Exercises

Exercise 15.1. Use the "multiples of the mean rule" to create a circular spinner for simulating values from a generic Exponential distribution.

1. Version 1: label the circular axis in increments of 0.5, 1.0, 1.5, 2.0, 2.5, etc., multiple of the mean. Careful: the values should not be evenly spaced.

2. Sketch a histogram with equal bin widths—0 to 0.5, 0.5 to 1.0, 1.0 to 1.5, etc.—that you would expect to see after spinning the version 1 spinner many times. Does it have an Exponential shape?

3. Version 2: label the circular axis so that your spinner has 10 sectors each with probability 0.1.

4. Sketch a histogram that you would expect to see after spinning the version 2 spinner many times. Start by sketching a histogram with *unequal* bin widths, then sketch the correspond equal bin width histogram. Does it have an Exponential shape?

5. Explain how could you use this spinner (either version) to simulate values from an Exponential distribution with rate parameter 2.5 (like in Example 15.1).

Exercise 15.2. Continuing Example 15.1. Let X be the waiting time (hours) until the next order and assume X has an Exponential(2.5) distribution.

1. Find the conditional probability that the waiting time from now until the next order is greater than 3 hours given that no orders occur in the next hour. Be sure to write a valid probability statement involving X before computing.

2. Compare to $\mathrm{P}(X > 2)$. What do you notice?

- **Memoryless property.** If W has an Exponential(λ) distribution then for any $w, h > 0$

$$\mathrm{P}(W > w + h \,|\, W > w) = \mathrm{P}(W > h)$$

- Given that we have already waited w units of time, the conditional probability that we wait at least an additional h units of time is the same as the unconditional probability that we wait at least h units of time.
- In this sense, the future waiting time does not "remember" how long we have already waited.
- A continuous random variable W has the memoryless property if and only if W has an Exponential distribution. That is, Exponential distributions are the *only* continuous[1] distributions with the memoryless property.
- If W has an Exponential(λ) distribution then the conditional distribution of the *excess* waiting time, $W - w$, given $\{W > w\}$ is Exponential(λ).

[1]Geometric distributions are the only discrete distributions with the discrete analog of the memoryless property.

16 Normal Distributions

- Normal distributions are probably the most important distributions in probability and statistics.
- A continuous random variable X has a **Normal** (a.k.a., Gaussian) distribution with mean $\mu \in (-\infty, \infty)$ and standard deviation $\sigma > 0$ if its pdf is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad -\infty < x < \infty$$

- If X has a Normal(μ, σ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \mu \\ \mathrm{SD}(X) &= \sigma \end{aligned}$$

- A Normal density is a particular "bell-shaped" curve which is symmetric about its mean μ. The mean μ is a *location* parameter: μ indicates where the center and peak of the distribution is.
- The standard deviation σ is a *scale* parameter: σ indicates the distance from the mean to where the concavity of the density changes. That is, there are inflection points at $\mu \pm \sigma$.

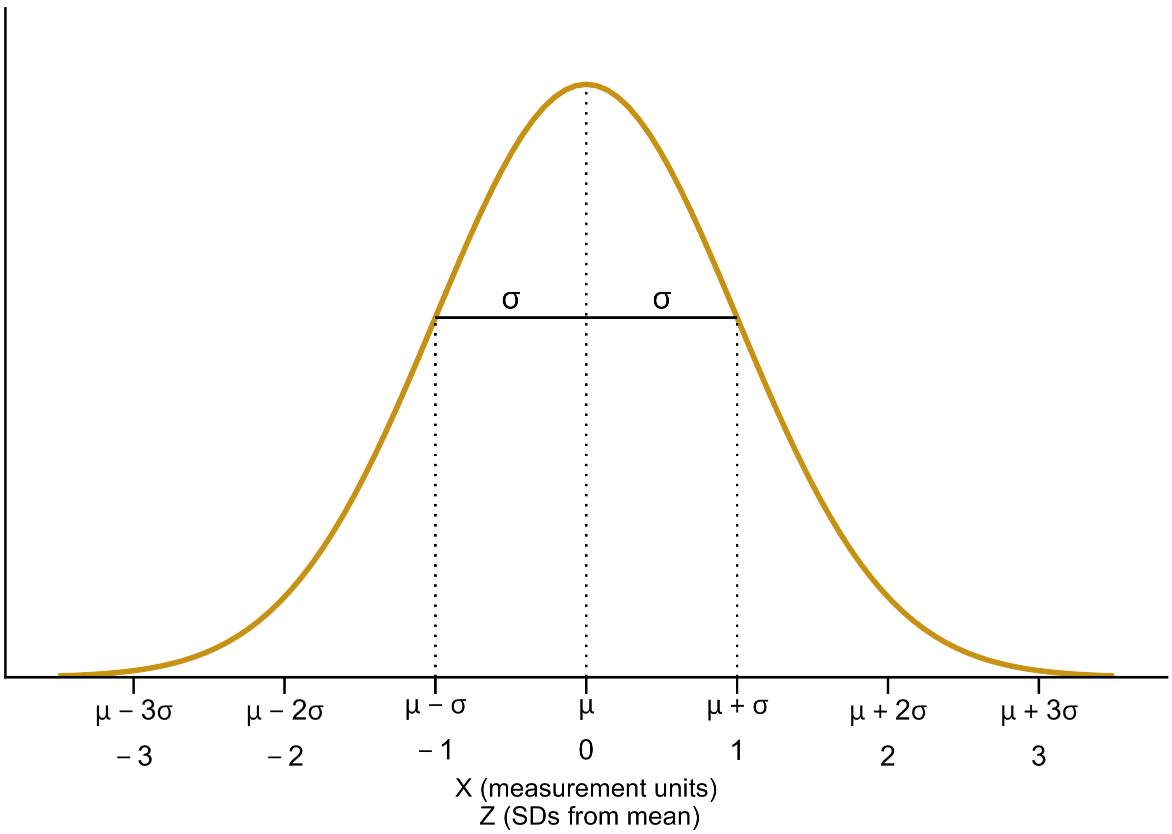


Figure 16.1: A generic Normal distribution

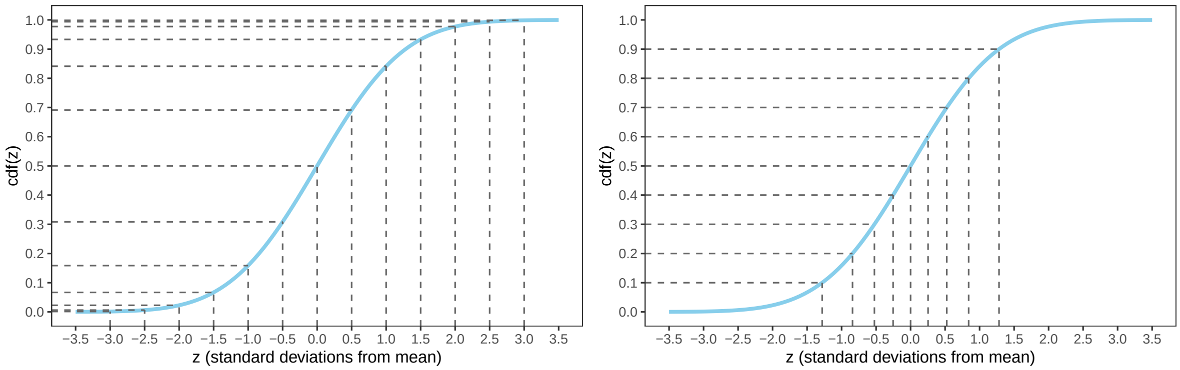
- The key to working with Normal distributions is to work in terms of standard deviations from the mean (that is, standardized values).
  - If X has a Normal(μ, σ) distribution then $Z = \frac{X-\mu}{\sigma}$ has a Normal(0, 1) distribution
  - If Z has a Normal(0, 1) distribution then $X = \mu + \sigma Z$ has a Normal(μ, σ) distribution
- Any Normal distribution follows an "empirical rule" which relates percentiles to standard deviation from the mean and defines the particular bell shape.
- A continuous random variable Z has a **Standard Normal** (a.k.a. Normal(0, 1)) distribution if its pdf is

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2}, \quad -\infty < z < \infty,$$

- The cdf is a Normal(0, 1) distribution is

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}}\, e^{-u^2/2}, \quad -\infty < z < \infty,$$

  but integrals of this type do not have a closed form and must be evaluated with software



(a) Percentiles highlighted in 0.5 increments of standard deviations from the mean

(b) Deciles highligted in terms of standard deviations from the mean

Figure 16.2: The Normal(0, 1) cdf. Provides the cdf for any Normal distribution if the variable is measured in *standard deviations from the mean*. It's the same cdf in both figures just with different values highlighted.

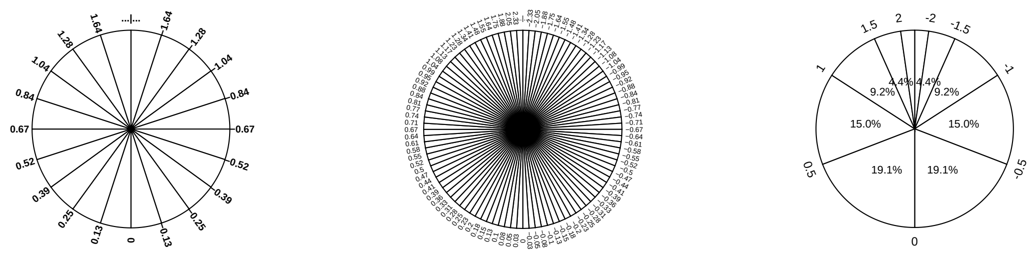
Table 16.1: Select percentiles for a Normal distribution in terms of standard deviations from the mean

(a) In 0.5 increments of standard deviations from the mean

Percentile	Standard deviations from the mean
0.02%	-3.5
0.1%	-3.0
0.6%	-2.5
2.3%	-2.0
6.7%	-1.5
15.9%	-1.0
30.9%	-0.5
50%	0.0
69.1%	0.5
84.1%	1.0
93.3%	1.5
97.7%	2.0
99.4%	2.5
99.9%	3.0
99.98%	3.5

(b) Deciles and quantiles as standard deviations from the mean

Percentile	Standard deviations from the mean
1%	-2.33
5%	-1.64
10%	-1.28
20%	-0.84
25%	-0.67
30%	-0.52
40%	-0.25
50%	0.00
60%	0.25
70%	0.52
75%	0.67
80%	0.84
90%	1.28
95%	1.64
99%	2.33



(a) Each sector represents an interval with 5% probability

(b) Each sector represents an interval with 1% probability

(c) Values labeled in increments of 0.5 standard deviations from the mean

Figure 16.3: A continuous *Normal(0, 1)* spinner. The same spinner is displayed in all figures, with different features highlighted. This spinner can be used to simulate values from any Normal distribution by simulating values in terms of standard deviations from the mean.

- Figure 16.2 defines a Normal distribution by specifying all of its percentiles.
- The empirical rule is often described in terms of "within [blank] standard deviations of the mean" as in the following:
  - 38% of values are within 0.5 standard deviations of the mean
  - 50% of values are withing 0.67 standard deviations of the mean
  - 68% of values are within 1 standard deviation of the mean
  - 87% of values are within 1.5 standard deviations of the mean
  - 95% of values are within 2 standard deviations of the mean
  - 99% of values are within 2.6 standard deviations of the mean
  - 99.7% of values are within 3 standard deviations of the mean
  - 99.99% of values are within 4 standard deviations of the mean

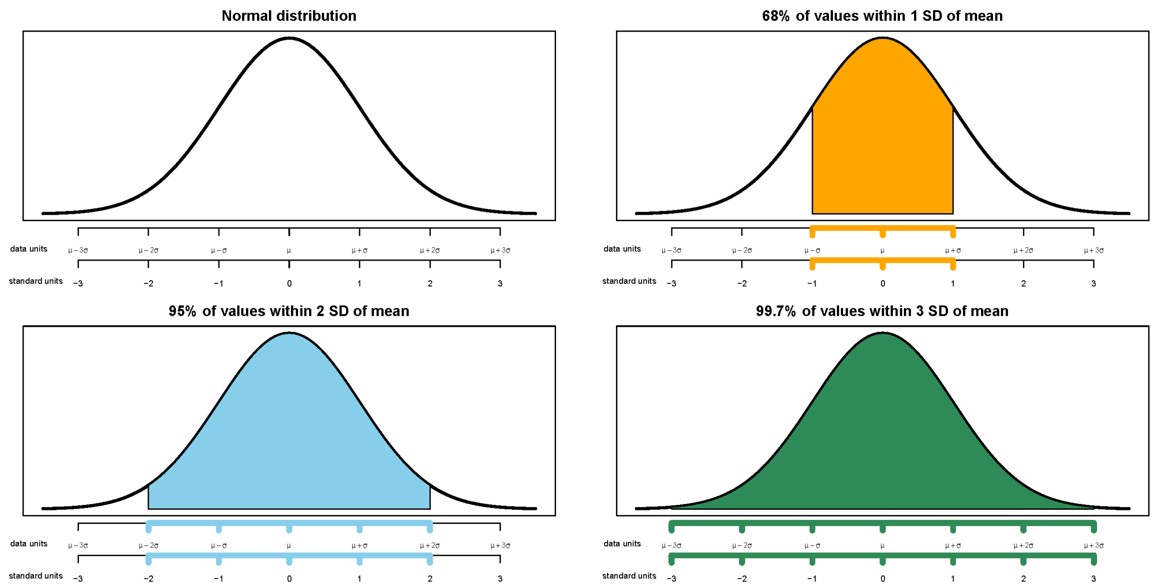


Figure 16.4: Illustrations of the empirical rule for Normal distributions

Example 16.1. The wrapper of a package of candy lists a weight of 47.9 grams. Naturally, the weights of individual packages vary somewhat. Suppose package weights have an approximate Normal distribution with a mean of 49.8 grams and a standard deviation of 1.3 grams.

1. Sketch the distribution of package weights. Carefully label the variable axis. It is helpful to draw two axes: one in the measurement units of the variable, and one in standardized units.

2. Why wouldn't the company print the mean weight of 49.8 grams as the weight on the package?

3. Estimate $\mathrm{P}(X < 47.9)$ and interpret the value in context.

4. Estimate P($47.9 < X < 53.0$) and interpret the value in context.

5. Suppose that the company only wants 1% of packages to be underweight. Find the weight that must be printed on the packages.

6. Find the 99th percentile of package weights.

16.1 Exercises

Exercise 16.1. A bank uses an applicant's score on some criteria to decide whether or not to approve their loan application. Based on past history, the bank has determined that

- Scores for those who repay the loan follow a Normal distribution with mean 60 and SD 8
- Scores for those who do NOT repay the loan follow a Normal distribution with mean 40 and SD 12

When someone applies for a loan the bank does not know whether the applicant will eventually repay the loan. How can the bank use the applicant's score to decide whether or not to approve the loan?

1. Draw sketches of these two normal curves on the same axis.

2. Suggest a general form for the decision rule, based on an applicant's score, for deciding whether or not to approve the applicant's loan. ("Approve the loan if the score…") Just give the general idea; you're not computing anything yet.

3. Describe in words the two kinds of errors the bank could make in this situation.

4. Determine the probability that the bank incorrectly rejects the application of someone who would repay. Shade in the corresponding region under the Normal curve, and interpret this probability.

5. Determine the probability that the bank incorrectly approves the application of someone who would NOT repay. Shade in the corresponding region under the Normal curve, and interpret this probability.

6. In which direction—smaller or larger—would you need to change the decision rule's cutoff value in order to decrease the probability that an applicant who would repay the loan is incorrectly rejected? Would the probability of the other kind of error—incorrectly approving a loan for an applicant who would not repay it—increase or decrease with this new cutoff value?

7. Determine the cutoff value needed to decrease the probability that an applicant who would repay the loan is incorrectly rejected to 0.05.

8. Determine the other error probability with this new cut-off rule.

9. Now repeat the two previous parts with the goal of decreasing to 0.05 the probability of incorrectly approving an applicant who would not repay the loan.

10. If you consider the two kinds of errors to be equally serious, how might you decide which of the three decision rules considered so far is the best?

11. Which error do you think the bank would consider more serious? In which direction would that move the threshold?

- Making decisions/predictions based on data often involves trade-offs.
- Decreasing the probability of one kind of error often comes at the expense of increasing the probability of another kind of error.

17 Joint and Conditional Distributions

- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) *pairs*, and describes how the values of X and Y vary together or jointly.
- Marginal distributions can be obtained from a joint distribution by "stacking"/"collapsing"/"aggregating" out the other variable.
- **In general, marginal distributions alone are not enough to determine a joint distribution.** (The exception is when random variables are *independent*.)

Example 17.1. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Recall Table 6.1.

1. Compute and interpret $p_{X,Y}(5, 3) = \mathrm{P}(X = 5, Y = 3)$.

2. Construct a "flat" table displaying the distribution of (X, Y) pairs, with one pair in each row.

3. Construct a two-way displaying the joint distribution on X and Y.

Table 17.1: Two-way table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

$x \setminus y$	1	2	3	4
2	1/16	0	0	0

3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

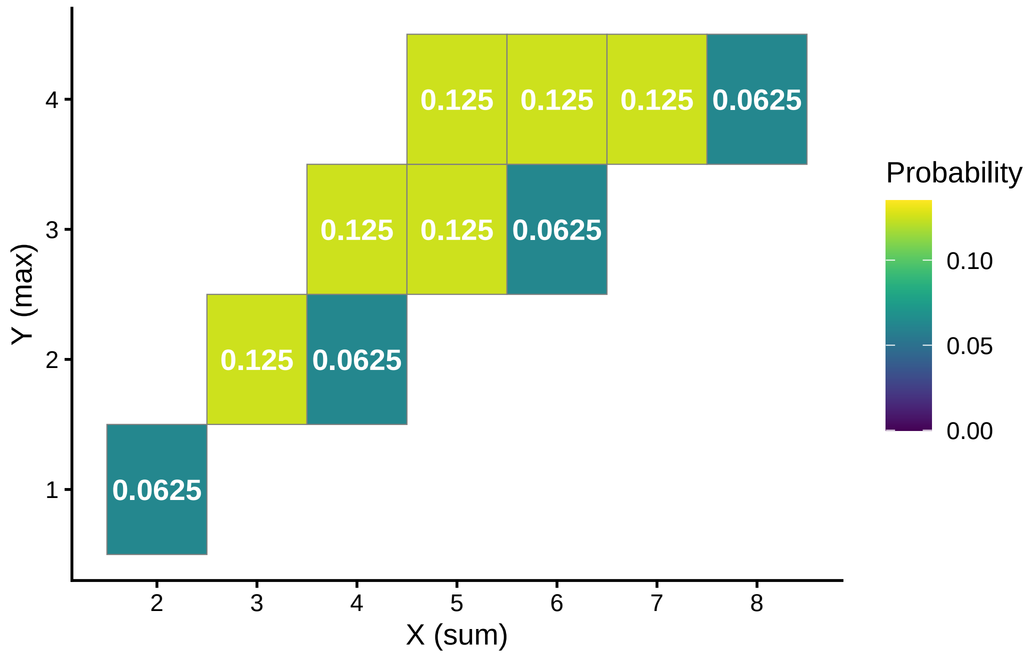


Figure 17.1: Tile plot representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

- The **conditional distribution of** Y given $X = x$ is the distribution of Y values over only those outcomes for which $X = x$. It is a distribution on values of Y only; treat x as a fixed constant when conditioning on the event $\{X = x\}$.
- Conditional distributions can be obtained from a joint distribution by *slicing and renormalizing*. The conditional distribution of Y given $X = x$, where x represents a particular number, can be thought of as:
  - the slice of the joint distribution corresponding to $X = x$, a distribution on values of Y alone with $X = x$ fixed
  - renormalized so that the slice accounts for 100% of the probability over the values of Y

- The *shape* of the conditional distribution of Y given $X =$ is determined by the shape of the slice of the joint distribution over values of Y for the fixed x.
- For each fixed x, the conditional distribution of Y given $X = x$ is a different distribution on values of the random variable Y. There is not one "conditional distribution of Y given X", but rather a *family of conditional distributions of* Y given different values of X.
- Each conditional distribution is a distribution, so we can summarize its characteristics like mean and standard deviation. The conditional mean and standard deviation of Y given $X = x$ represent, respectively, the long run average and variability of values of Y over only (X, Y) pairs with $X = x$.
- Since each value of x typically corresponds to a different conditional distribution of Y given $X = x$, the conditional mean and standard deviation will typically be functions of x.

Example 17.2. Continuing Example 17.1.

1. Make a table representing the conditional distribution of X given $Y = 4$.

2. Make a table representing the conditional distribution of X given $Y = 3$.

3. Make a table representing the conditional distribution of X given $Y = 2$.

4. Make a table representing the conditional distribution of X given $Y = 1$.

- The **conditional expected value** (a.k.a. *conditional expectation* a.k.a. *conditional mean*), of a random variable Y given the event $\{X = x\}$, defined on a probability space with measure P, is a *number* denoted $\mathrm{E}(Y|X = x)$ representing the probability-weighted

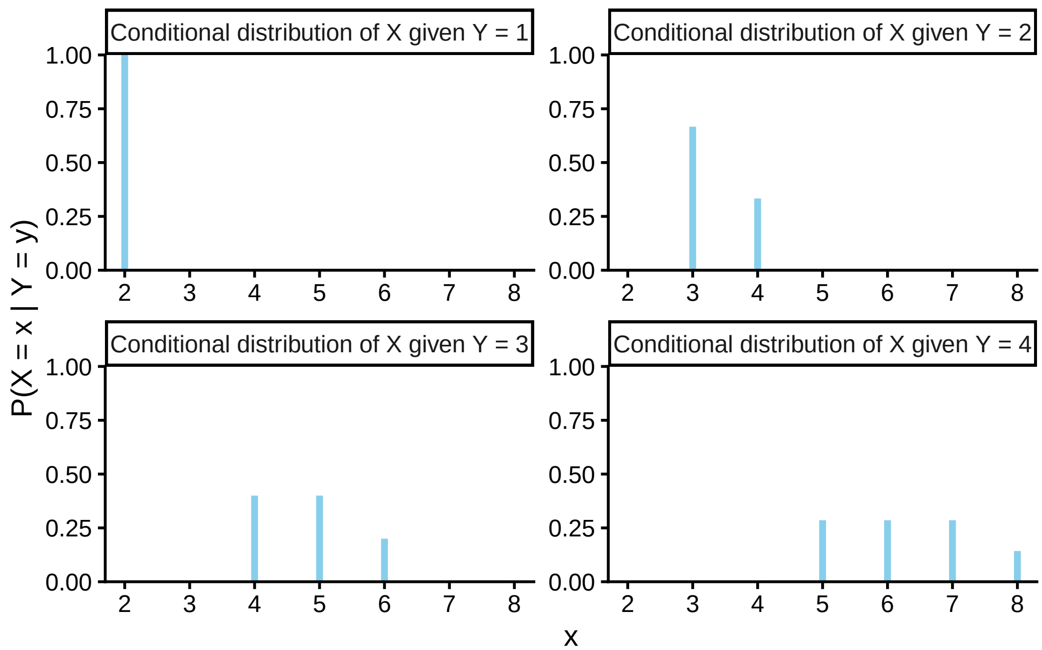


Figure 17.2: Impulse plots representing the family of conditional distributions of X given Y for the dice rolling example. Each plot represents a conditional distribution of X given $Y = y$ for a particular value of $y = 1, 2, 3, 4$.

average value of Y, where the weights are determined by the conditional distribution of Y given $X = x$.

$$\text{Discrete } X, Y \text{ with conditional pmf } p_{Y|X}: \qquad \mathrm{E}(Y|X = x) = \sum_y y p_{Y|X}(y|x)$$

- Remember, when conditioning on $X = x$, x is treated as a fixed constant. The conditional expected value $\mathrm{E}(Y|X = x)$ is a *number* representing the mean of the conditional distribution of Y given $X = x$.
- The conditional expected value $\mathrm{E}(Y|X = x)$ is the long run average value of Y over only those outcomes for which $X = x$.
- To approximate $\mathrm{E}(Y|X = x)$, simulate many (X, Y) pairs, discard the pairs for which $X \neq x$, and average the Y values for the pairs that remain.

Example 17.3. Continuing Example 17.2.

1. Compute and interpret $\mathrm{E}(X|Y = 4)$.

2. Without computing, will $\mathrm{E}(X|Y = 3)$ be greater than, less than, or equal to $\mathrm{E}(X|Y = 4)$?

- The **joint probability mass function (pmf)** of two *discrete* random variables (X, Y) defined on a probability space is the function defined by

$$p_{X,Y}(x, y) = \mathrm{P}(X = x, Y = y) \qquad \text{for all } x, y$$

- Remember to specify the possible (x, y) pairs when defining a joint pmf.

Example 17.4. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y have joint pmf

$$p_{X,Y}(x, y) = \begin{cases} e^{-2.4} \frac{1.21^x 1.19^y}{x!y!}, & x = 0, 1, 2, \dots; y = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

1. Compute and interpret $\mathrm{P}(X = 2, Y = 1)$.

2. Construct a two-way table representation of the joint pmf (you can use software or a spreadsheet).

3. Compute and interpret $\mathrm{P}(X \leq 3, Y \leq 3)$.

4. Compute and interpret $\mathrm{P}(X + Y = 3)$.

5. Compute and interpret $\mathrm{P}(X = Y)$.

6. Compute and interpret $\mathrm{P}(X = 2)$.

7. Compute and interpret $\mathrm{P}(Y = 1)$.

- Two random variables X and Y defined on a probability space with probability measure P are **independent** if $\mathrm{P}(X \le x, Y \le y) = \mathrm{P}(X \le x)\mathrm{P}(Y \le y)$ for all x, y. That is, two random variables are independent if their joint cdf is the product of their marginal cdfs.
- Random variables X and Y are independent if and only if the joint distribution factors into the product of the marginal distributions. The definition is in terms of cdfs, but analogous statements are true for pmfs and pdfs.

$$\begin{aligned} &\text{Discrete RVs } X \text{ and } Y \text{ are independent} \\ &\qquad \Longleftrightarrow p_{X,Y}(x,y) = p_X(x)p_Y(y) \qquad\qquad \text{for all } x, y \end{aligned}$$

$$\begin{aligned} &\text{Continuous RVs } X \text{ and } Y \text{ are independent} \\ &\qquad \Longleftrightarrow f_{X,Y}(x,y) = f_X(x)f_Y(y) \qquad\qquad \text{for all } x, y \end{aligned}$$

17.1 Exercises

Exercise 17.1. Continuing Example 17.1.

1. Starting with the two-way table representing the joint distribution of X and Y, how could you obtain $\mathrm{P}(X = 5)$?

2. Starting with the two-way table, how could you obtain the marginal distribution of X? of Y?

3. Starting with the marginal distribution of X and the marginal distribution of Y, could you necessarily construct the two-way table of the joint distribution? Explain.

4. Remember that you constructed a spinner corresponding to the marginal distribution of Y in Example 6.1. Suppose you also construct a separate spinner corresponding to the marginal distribution of X. Donny Don't says: "to simulate an (X, Y) pair I can just

spin the X spinner once to get X and spin the Y spinner once to get Y." Do you agree? Explain.

- Recall that we can obtain marginal distributions from a joint distribution.
- *Marginal pmfs* are determined by the joint pmf via the law of total probability.
- If we imagine a plot with blocks whose heights represent the joint probabilities, the marginal probability of a particular value of one variable can be obtained by "stacking" all the blocks corresponding to that value.
- For discrete random variables X and Y:

$$\begin{aligned} p_X(x) &= \sum_y p_{X,Y}(x,y) && \text{a function of } x \text{ only} \\ p_Y(y) &= \sum_x p_{X,Y}(x,y) && \text{a function of } y \text{ only} \end{aligned}$$

Exercise 17.2. Continuing Exercise 17.1.

1. Construct a spinner that could be used to simulate (X, Y) pairs with the proper joint distribution.

2. How could you use Example 17.2 to construct a series of spinners to simulate (X, Y) pairs with the proper joint distribution? (Hint: consider simulating Y first; then how would you simulate X?)

3. Explain how you could use simulation to approximate the conditional distribution of Y given $X = 6$.

4. Explain how you could use simulation to approximate $\mathrm{E}(Y|X = 6)$.

- Rather than directly simulating from a joint distribution, we can simulate an (X, Y) pair in two stages:
  - Simulate a value of X from its marginal distribution. Call the simulated value x.
  - Given x, simulate a value of Y from the conditional distribution of Y given $X = x$. There will be a different distribution (spinner) for each possible value of x.
- This "marginal then conditional" process is essentially implementing the multiplication rule

$$\text{joint} = \text{conditional} \times \text{marginal}$$

- In many problems a joint distribution is naturally described by specifying the marginal distribution of X and the family of conditional distributions of Y given values of X. (This is sometimes called a "hierarchical model".)

18 Covariance and Correlation

Quantities like expected value and variance summarize characteristics of the *marginal* distribution of a single random variable. When there are multiple random variables their *joint* distribution is of interest. *Covariance* and *correlation* summarize a characteristic of the joint distribution of two random variables, namely, the degree to which they "co-deviate from the their respective means".

Example 18.1. We'll motivate the calculation of covariance by considering a small example. Say we have data on the sale price X (measured in millions of dollars) and the living area Y (measured in thousands of square feet) for a small sample of 10 houses.

repetition	X	Y	X deviation	Y deviation	Product
1	0.28	1.12			
2	0.63	1.05			
3	0.76	1.67			
4	0.56	1.48			
5	0.36	0.43			
6	0.89	2.50			
7	0.47	1.71			
8	0.82	1.88			
9	0.71	0.85			
10	0.75	1.90			
Sum	6.23	14.59			
Average	0.62	1.46			

1. Compute the average of the X values and the average of the Y values.
2. Compute the deviation from its mean for each X value, and the deviation from its mean for each Y value.
3. Compute the product of the paired deviations.
4. Compute the average of the products of the deviations. This is called the covariance. What are the measurement units?

- The **covariance** between two random variables X and Y is

$$\begin{aligned}\text{Cov}(X,Y) &= \text{E}\left[(X-\text{E}[X])\,(Y-\text{E}[Y])\right]\\ &= \text{E}(XY)-\text{E}(X)\text{E}(Y)\end{aligned}$$

- Covariance is the long run average of the products of the paired deviations from the mean
- Covariance is the expected value of the product minus the product of expected values.

Example 18.2. Roll a fair four-sided die twice and let X be the sum of the two rolls, and Y the larger of the two rolls. Recall Example 17.1 and Table 17.1.

1. Will $\text{Cov}(X,Y)$ be positive, negative, or close to 0? Explain without computing.

2. Compute $\text{Cov}(X,Y)$ using the definition.

3. Compute $\text{Cov}(X,Y)$ using the "shortcut" formula.

- $\text{Cov}(X,Y) > 0$ (positive association): above average values of X tend to be associated with above average values of Y
- $\text{Cov}(X,Y) < 0$ (negative association): above average values of X tend to be associated with below average values of Y
- $\text{Cov}(X,Y) = 0$ indicates that the random variables are **uncorrelated**: there is no overall positive or negative association. But be careful: if X and Y are uncorrelated there can still be a relationship between X and Y; there is just no overall positive or negative association.. We will see examples that demonstrate that being uncorrelated does not necessarily imply that random variables are independent.

Example 18.3. We'll motivate the calculation of correlation by considering a small example. Continuing Example 18.1, say we have data on the sale price X (measured in millions of dollars) and the living area Y (measured in thousands of square feet) for a small sample of 10 houses.

repetition	X	Y	Standardized X	Standardized Y	Product
1	0.28	1.12			
2	0.63	1.05			
3	0.76	1.67			
4	0.56	1.48			
5	0.36	0.43			
6	0.89	2.50			
7	0.47	1.71			
8	0.82	1.88			
9	0.71	0.85			
10	0.75	1.90			
Sum	6.23	14.59	0	0	
Average	0.62	1.46	0	0	
SD	0.19	0.57	1	1	NA

1. Compute the average and SD of the X values and the average and SD of the Y values.
2. Compute the standardized value of each X value, and the standardized value of each Y value.
3. Compute the product of the paired standardized values.
4. Compute the average of the products of the standardized values. This is called the correlation.
5. Divide the covariance from Example 18.1 by the product of the two standard deviations. What do you notice?

- The **correlation (coefficient)** between random variables X and Y is

$$
\begin{aligned}
\text{Corr}(X, Y) &= \text{Cov}\left(\frac{X - \text{E}(X)}{\text{SD}(X)}, \frac{Y - \text{E}(Y)}{\text{SD}(Y)}\right) \\
&= \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}
\end{aligned}
$$

- The correlation for two random variables is the covariance between the corresponding standardized random variables. Therefore, correlation is a *standardized* measure of the association between two random variables.

- A correlation coefficient has no units and is measured on a universal scale. Regardless of the original measurement units of the random variables X and Y

$$-1 \leq \text{Corr}(X, Y) \leq 1$$

- $\text{Corr}(X, Y) = 1$ if and only if $Y = aX + b$ for some $a > 0$
- $\text{Corr}(X, Y) = -1$ if and only if $Y = aX + b$ for some $a < 0$
- Therefore, correlation is a standardized measure of the strength of the *linear* association between two random variables.
- Covariance is the correlation times the product of the standard deviations.

$$\text{Cov}(X, Y) = \text{Corr}(X, Y)\text{SD}(X)\text{SD}(Y)$$

18.1 Exercises

Exercise 18.1. Continuing Example 18.2. Let X be the sum of the two rolls, Y the larger of the two rolls, W the number of rolls equal to 4, and Z the number of rolls equal to 1. You should explain your answers without doing any computation.

1. Is $\text{Cov}(X, W)$ positive, negative, or zero? Why?

2. Is $\text{Cov}(X, Z)$ positive, negative, or zero? Why?

3. Let $V = W + Z$; what does this represent? Is $\text{Cov}(X, V)$ positive, negative, or zero? Why?

4. Is $\text{Cov}(W, Z)$ positive, negative, or zero? Why?

Exercise 18.2. What is another name for $\text{Cov}(X, X)$?

Exercise 18.3. Sketch a few scatterplots where X and Y have a correlation of 0, but they are not independent.

Exercise 18.4. Try the Guessing Correlations applet! Since we're interested in the long run, set number of points to 1000. There is nothing to submit for this exercise but play around with the applet to get a feel for the difference between a correlation of 0.8-ish versus 0.5-ish versus 0.2-ish.

Homework 1

1. In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)

    a. The probability of rolling doubles when you roll two fair six-sided dice is 1/6.

    b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.

    c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.

    d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.

2. Suppose that at some point in 2023 your subjective probabilities for who would win the 2024 U.S. Presidential Election satisfied the following. (Imagine a time before Trump had been declared the Republican nominee, and while Biden was the presumptive Democratic nominee there was still speculation that he should step aside.)

    - The Democratic nominee and the Republican nominee are equally likely to be the winner of the Presidential Election
    - Joe Biden is 5 times more likely to win than Kamala Harris, and no other Democratic candidate has a chance of winning
    - Donald Trump is twice as likely to win as any other Republican candidate.

    Create a table of your subjective probabilities.

3. Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

    Let A represent the event that you are accepted at school A, B the event that you are accepted at school B, and P the probability measure that represents your subjective probabilities. In addition to computing, represent each of the probabilities below with appropriate notation.

a. Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?
b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
f. If you are accepted at school A, what is your probability of not being accepted at school B?
g. If you are not accepted at school A, what is your probability of being accepted at school B?
h. If you are accepted at school B, what is your probability of also being accepted at school A?
i. If you are not accepted at school B, what is your probability of being accepted at school A?

4. For each of the following, create your own "which of (a) or (b) is larger?" example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don't do any calculations.

a. Reverse the direction of conditioning (like the 6 foot tall and NBA example)
b. Compare conditional and unconditional probabilities (like man, versus man who is greater than 6 feet tall)
c. Two "particulars" that people might think are not equally likely but are (like HHHHH versus HTHTT).
d. Comparing "the particular" and "the general" (HTHTT versus 2 H in 5 flips).
e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)

5. The "matching problem" involves n distincts objects labeled $1, \ldots, n$ which are placed in n distinct boxes labeled $1, \ldots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We're interested in the number of matches (the number of objects for

which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

For example, suppose n people have a secret Santa gift exchange. Each person puts their name in a hat, the names are well shuffled, and everyone draws a name from the hat (to then purchase a gift for the selected person). Then C represents the event that at least one person draw their own name.

a. Describe in full detail how, in principle, you would physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $\mathrm{P}(C)$.
c. This applet conducts a simulation of the matching problem (in the context of babies in a hospital being mixed up and returned to parents uniforomly at random.) In the applet, "Number of babies" represents n and "Number of trials" represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the "Randomize" button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $\mathrm{P}(C)$ when $n = 4$.
d. How do you expect $\mathrm{P}(C)$ to depend on n? As n increases, do you expect $\mathrm{P}(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you're free to choose other values). For each value of n run the simulation and use the results to approximate $\mathrm{P}(C)$. Record your results.
f. What do the simulation results suggest about the relationship between $\mathrm{P}(C)$ and n? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margion of error of each other are essentially the same approximation.)

6. Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss's dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

   a. Compute the probability that Katniss's dart lands within 1 inch of the center of the dartboard.
   b. Compute the probability that Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

7. Imagine a light that flashes every few seconds[1]. The light randomly flashes green with probability 0.75 and red with probability 0.25, independently from flash to flash.

[1] Thanks to Allan Rossman for this example.

a. Write down a sequence of G's (for green) and R's (for red) to predict the colors for the next 40 flashes of this light. Before you read on, please take a minute to think about how you would generate such a sequence yourself.
b. Most people produce a sequence that has 30 G's and 10 R's, or close to those proportions, because they are trying to generate a sequence for which each outcome has a 75% chance for G and a 25% chance for R. That is, they use a strategy in which they predict G with probability 0.75, and R with probability 0.25. How well does this strategy do? Compute the probability of correctly predicting any single item in the sequence using this strategy.
c. Describe a better strategy. (Hint: can you find a strategy for which the probability of correctly predicting any single flash is 0.75?)

Homework 2

Problem 1

A standard deck of playing cards has 52 cards, 13 cards (2 through 10, jack, king, queen, ace) in each of 4 suits (hearts, diamonds, clubs, spades). Shuffle a deck and deals cards one at a time without replacement.

1. What is the probability that the first card dealt is a heart?
2. What is the probability that the second card dealt is a heart? Just make a guess now before proceeding, but you'll revisit this a few parts down.
3. What is the probability that the second card dealt is a heart if the first card dealt is a heart?
4. What is the probability that the second card dealt is a heart if the first card dealt is not a heart?
5. Revisit part 2. What is the probability that the second card dealt is a heart? Hint: your answer should be a single number, but show all your work.
6. Find the probability that the first two cards dealt are hearts.
7. Find the probability that the first two cards dealt are hearts and the third card dealt is a diamond.

Problem 2

The ELISA test for HIV was widely used in the mid-1990s for screening blood donations. As with most medical diagnostic tests, the ELISA test is not perfect. If a person actually carries the HIV virus, experts estimate that this test gives a positive result 97.7% of the time. (This number is called the *sensitivity* of the test.) If a person does not carry the HIV virus, ELISA gives a negative (correct) result 92.6% of the time (the *specificity* of the test). Estimates at the time were that 0.5% of the American public carried the HIV virus.

Suppose that a randomly selected American tests positive; we are interested in the conditional probability that the person actually carries the virus.

1. Before proceeding, make a guess for the probability in question. Which range do you think it's in: 0-20%, 20-40%, 40-60%, 60-80%, or 80-100%?
2. Denote the probabilities provided in the setup using proper notation

3. Construct an appropriate two-way table and use it to compute the probability of interest.
4. Construct a Bayes table and use it to compute the probability of interest.
5. Explain why this probability is small, compared to the sensitivity and specificity.
6. By what factor has the probability of carrying HIV increased, given a positive test result, as compared to before the test?
7. Now suppose that 5% of individuals in a high-risk group carry the HIV virus. Consider a randomly selectd person from this group who takes the test. Given that the test is positive, how many times more likely is it for the person not to have HIV than to have it? Answer without first computing a two-way or Bayes table.
8. Using the result from the previous part, compute the conditional probability that a person in this risk group who tests positive has HIV.
9. Is the posterior probability influenced by the prior probability? Discuss.

Problem 3

Consider three tennis players Arya ("A"), Brienne ("B"), and Cersei ("C"). One of these players is better than the other two, who are equally good/bad. When the best player plays either of the others, she has a 2/3 probability of winning the match. When the other two players play each other, each has a 1/2 probability of winning the match. But you do not know which player is the best. Based on watching the players warm up, you start with subjective probabilities of 0.5 that A is the best, 0.35 that B is the best, and 0.15 that C is the best. A and B will play the first match.

1. Suppose that A beats B in the first match. Compute your posterior probability that each of A, B, C is best given that A beats B in the first match.
2. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
3. **Coding required.** Code and run a simulation to simulate (1) who is the best player according to your prior distribution, (2) who wins the first match given who is the best player; then use the results to approximate the probabilities from the first part.
4. Suppose instead that B beats A in the first match. Compute your posterior probability that each of A, B, C is best given that B beats A in the first match.
5. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
6. Now suppose again that A beats B in the first match, and also that A beats C in the second match. Compute your posterior probability that each of A, B, C is best given the results of the first two matches. (Hint: use as the prior your posterior probabilities from the previous part.) Explain how your probabilities changed, and why that makes sense.

Problem 4

In the Powerball lottery, a player picks five different whole numbers between 1 and 69, and another whole number between 1 and 26 that is called the Powerball. In the drawing, the 5 numbers are drawn without replacement from a "hopper" with balls labeled 1 through 69, but the Powerball is drawn from a separate hopper with balls labeled 1 through 26. The player wins the jackpot if both the first 5 numbers match those drawn, in any order, and the Powerball is a match. Under this set up, there are 292,201,338 possible winning numbers.

1. What is the probability the next winning number is 6-7-16-23-26, plus the Powerball number, 4.
2. What is the probability the next winning number is 1-2-3-4-5, plus the Powerball number, 6.
3. The Powerball drawing happens twice a week. Suppose you play the same Powerball number, twice a week, every week for over 50 years. Let's say you purchase a ticket for 6000 drawings in total. What is the probability that you win at least once?
4. Instead of playing for 50 years, you decide only to play one lottery, but you buy 6000 tickets, each with a different Powerball number. What is the probability that at least one of your tickets wins? How does this compare to the previous part? Why?
5. Each ticket costs 2 dollars, but the jackpot changes from drawing to drawing. Suppose you buy 6000 tickets for a single drawing. How large does the jackpot need to be for your "expected" profit to be positive? To be $100,000? (We're ignoring inflation, taxes, transaction costs, and any changes in the rules.)

Problem 5

Consider a "best-of-5" series of games between two teams: games are played until one of the teams has won 3 games (requiring at most 5 games total). Suppose one team, team A, is better than the other, having a 0.55 probability of winning any particular game. Assume the results of the games are independent (and ignore advantage, etc). Let X represent the number of games played in the series. (Hint: It's helpful to first construct a two-way table of probabilities with the number of games played and which team wins, and then use it to answer the following questions. It will also help to list some outcomes, like AABA (team A wins game 1, 2, and 4, and B wins game 3).)

1. Compute the probability that team A wins the series in 3 games.
2. Compute the probability that the series ends in 3 games.
3. Compute the probability that team A wins the series.
4. Are the events "team A wins the series" and "the series ends in 3 games" independent? Explain by comparing relevant probabilities.
5. Let X represent the number of games played in the series. Find the distribution of X.

Homework 3

Problem 1

Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that at the end of every practice session, she attempts three pointers until she makes one and then stops. Let X be the total number of shots she attempts in a practice session. Assume shot attempts are independent, each with probability of 0.4 of being successful.

1. Identify by name the distribution of X. Be sure to specify the values of any relevant parameters.
2. Specify the probability mass function of X. Be sure to specify the possible values.
3. Construct a table, plot, and spinner corresponding to the distribution of X.
4. Compute $\mathrm{P}(X > 5)$ without summing.
5. Compute and interpret $\mathrm{E}(X)$.
6. Compute and interpret $\mathrm{SD}(X)$.

Problem 2

Ron and Leslie agree to the following bet. They'll ask Professor Ross if he saw the Eras Tour live. If he did, Leslie will pay Ron \$200; if not, Ron will pay Leslie \$100. (Neither has any direct information about whether or not Prof Ross saw the Eras Tour live.)

1. Given this setup, which of the following is being judged as more likely: that Prof Ross saw the Eras Tour, or that he did not? Why?
2. Ron and Leslie agree that this is a fair bet, and neither would accept worse odds. What is their subjective probability that Professor Ross saw the Eras Tour?
3. Suppose they were to hypothetically repeat this bet many times, say 3000 times. Given the probability from the previous part, how many times would you expect Leslie to win? To lose? What would you expect Leslie's net dollar winnings to be? In what sense is this bet "fair"?
4. Let X be Leslie's net winnings. Identify the distribution of X. (Remember: Leslie's winnings are Ron's losses and vice versa.)
5. Compute and interpret $\mathrm{E}(X)$.

Problem 3

Suppose that a total of 350 students at a college are taking a particular statistics course. The college offers five sections of the course, each taught by a different instructor. The class sizes are shown in the following table.

Section	A	B	C	D	E
Number of students	35	35	35	35	210

We are interested in: What is the average class size?

1. Suppose we randomly select one of the 5 *instructors*. Let X be the class size for the selected instructor. Specify the distribution of X. (A table is fine.)
2. Compute and interpret $\mathrm{E}(X)$.
3. Compute and interpret $\mathrm{P}(X = \mathrm{E}(X))$.
4. Suppose we randomly select one of the 350 *students*. Let Y be the class size for the selected student. Specify the distribution of Y. (A table is fine.)
5. Compute and interpret $\mathrm{E}(Y)$.
6. Compute and interpret $\mathrm{P}(Y = \mathrm{E}(Y))$.
7. Comment on how these two expected values compare, and explain why they differ as they do. Which average would you say is more relevant?

Problem 4

(This is an overly simplified example from actuarial science.) A life insurance company sells a term insurance policy to a 21 year old man (the "insured") that pays \$100,000 to a designated beneficiary if the insured dies within the next 5 years, and pays nothing if the insured does not die before age 26. The probability that a randomly chosen 21-year-old man will die each year can be found in mortality tables. The company collects a premium of \$250 at the beginning of each year of the 5 years of the term, as long as the insured is alive, as payment for the insurance. The amount Y (in dollars) that the company earns on this policy is \$250 per year, less the \$100,000 that it pays out in the event the insured dies. The company doesn't pay out anything if the insured does not die in the 5 year term.

Age at death	Earnings Y	Probability
21	$-99{,}750$	0.00183
22		0.00186
23		0.00189
24		0.00191
25		0.00193

Age at death	Earnings Y	Probability
26 or older		

1. Find the distribution of Y by completing the above table. For example, the value -99,750 provides an example of how Y is calculated when the insured dies in the first year of the policy.
2. Compute $\mathrm{E}(Y)$.
3. Write a sentence explaining in this context in what sense the number from the previous part is "expected".
4. Would you be willing to sell such an insurance policy to one of your 21 year old friends? Explain why this is good business for the insurance company, but not for you.
5. Compute $\mathrm{Var}(Y)$
6. Compute $\mathrm{SD}(Y)$.
7. Compute the probability that Y is more than 1 standard deviation above its mean.

Problem 5

In a sample of 50 Olympic women's gymnasts the heights have a symmetric bell-shaped distribution with mean 60 inches and standard deviation 3 inches. In a sample of 50 Olympic men's basketball players the heights have symmetric bell-shaped distribution with mean 76 inches and the standard deviation 5 inches. Suppose the two samples are combined to obtain a sample with 100 heights. Choose one of the options below to complete the following sentence; explain your reasoning without doing any calculations (hint: drawing a picture might help; also, what is the mean of the combined sample?). The standard deviation of the combined sample is:

a. Less than 3 inches
b. Equal to 3 inches
c. Between 3 and 5 inches
d. Equal to 5 inches
e. Greater than 5 inches

Homework 4

Problem 1

Xavier bets on red on each of 3 spins of a roulette wheel. After the third spin, Xavier leaves and Zander arrives and bets on black on the next two spins. Let X be the number of bets that Xavier wins and let Z be the number that Zander wins.

1. Identify the distribution of X and the distribution of Z.
2. Are X and Z independent? Explain.
3. Compute and interpret $\mathrm{P}(X = 2, Z = 1)$.
4. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations. Be sure to specify the values of any parameters. Explain your reasoning.
5. Compute $\mathrm{P}(X + Z = 3)$ without summing many terms.
6. Suppose instead that Yolanda bets on black on all 5 spins of the wheel. Does $X + Y$ have a Binomial distribution? Answer yes or no and explain.
7. Suppose instead that Zander bet on the number 7 in his two bets. Does $X + Z$ have a Binomial distribution in this case? Answer yes or no and explain.

Problem 2

Suppose the number of earthquakes per hour, for a certain range of magnitudes in a certain region, follows a Poisson distribution with parameter 0.7, independently from hour to hour.

1. Compute and interpret the probability that there is at least one earthquake of this size in the region in any given hour.
2. Compute and interpret the probability that there are exactly 3 earthquakes of this size in the region in any given hour.
3. Interpret the value 0.7 in context.
4. Construct a table, plot, and spinner corresponding to a Poisson(0.7) distribution.
5. Let X be the number of earthquakes in a 2 hour period. Identify by name the distribution of X. Be sure to specify the values of any relevant parameters.
6. Compute and interpret $\mathrm{P}(X = 3)$.

Problem 3

Recall Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60. Recall the sample space from Example 2.2 and assume a uniform probability measure P.

Let R be Regina's arrival time and Y Cady's. Then $W = |R - Y|$ is the amount of time (minutes) the first person to arrive waits for the second person to arrive.

1. Compute and interpret $\mathrm{P}(W < 15)$. (Hint: we computed this already in Example 2.6.)
2. Compute and interpret $\mathrm{P}(W > 45)$. (Hint: use a method similar to the previous part.)
3. It can be shown that W has pdf

$$f_W(w) = (2/3600)(1 - w), \qquad 0 < w < 60.$$

   Sketch a plot of the pdf. Describe roughly what this means in context.
4. Use the pdf to compute $\mathrm{P}(W < 15)$. Set up an integral, but sketch a picture and use geometry to compute.
5. Use the pdf to compute $\mathrm{P}(W > 45)$. Set up an integral, but sketch a picture and use geometry to compute.

Problem 4

Continuing with the setup of Problem 3. Consider $T = \min(R, Y)$ is the time (minutes after noon) at which the first person arrives.

1. Is T the same random variable as W from Problem 3? Explain.
2. Compute and interpret $\mathrm{P}(T < 15)$. (Hint: we computed something similar in Example 2.6.)
3. Compute and interpret $\mathrm{P}(T > 45)$. (Hint: use a method similar to the previous part.)
4. It can be shown that T has pdf

$$f_T(t) = (2/3600)(1 - t), \qquad 0 < t < 60.$$

   Does T have the same distribution as W from Problem 3? Explain what this means.

Homework 1 Solutions

Problem 1

In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)

a. The probability of rolling doubles when you roll two fair six-sided dice is 1/6.

b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.

c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.

d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.

Solution

a. In about 16.7% sets of two rolls both rolls in the set are 6.
b. In about 0.463% of sets of three rolls all three rolls are 6. (If the sets consisted of more rolls, say 10 rolls, the probability of at least 3 consecutive 6s somewhere in the set would be greater.)
c. In about 12% of sets of 100 rolls the sum of the 100 rolls is less than 370.
d. In about 82.2% of sets of 9 rolls there are at most 2 6s.

Problem 2

1. Suppose that at some point in 2023 your subjective probabilities for who would win the 2024 U.S. Presidential Election satisfied the following. (Imagine a time before Trump had been declared the Republican nominee, and while Biden was the presumptive Democratic nominee there was still speculation that he should step aside.)

- The Democratic nominee and the Republican nominee are equally likely to be the winner of the Presidential Election
- Joe Biden is 5 times more likely to win than Kamala Harris, and no other Democratic candidate has a chance of winning
- Donald Trump is twice as likely to win as any other Republican candidate.

Create a table of your subjective probabilities.

Solution

It helps to designate one outcome as the "baseline". It doesn't matter which one; we'll choose Kamala Harris.

- Suppose Harris accounts for 1 "unit".
- Biden accounts for 5 units
- Other Democrat candidates account for 0 units
- Democrats in total account for 6 units
- Republicans in total account for 6 units
- Trump accounts for 4 units
- Other Republican accounts for 2 units

Candidate	"Units"	Probability	As Percent
Biden	5	$5/12 = 0.417$	41.7%
Harris	1	$1/12 = 0.083$	8.3%
Other Democrat	0	0	0%
Trump	4	$4/12 = 0.333$	33.3%
Other Republican	2	$2/12 = 0.167$	16.7%
Total	12	1	100%

Problem 3

Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

a. Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?

b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
f. If you are accepted at school A, what is your probability of not being accepted at school B?
g. If you are not accepted at school A, what is your probability of being accepted at school B?
h. If you are accepted at school B, what is your probability of also being accepted at school A?
i. If you are not accepted at school B, what is your probability of being accepted at school A?

Solution

a. The probability of being accepted at both schools can be no more than the probability of being accepted at either school, 0.6 and 0.7. So we try 0.6 and see if we get valid probabilities.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.6	0.1	0.7
Not accepted by B	0	0.3	0.3
Total	0.6	0.4	1.00

All the probabilities above are valid, so the largest the probability of being accepted at both schools can be is 0.6. This occurs if getting into school A guarantees that you will also get into school B.

b. Trying a subjective probability of being accepted at both schools leads to a negative probability of not being accepted at either school. So try a probability of 0 for the probability of not being accepted at either school.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.3	0.4	0.7
Not accepted by B	0.3	0	0.3

	Accepted by A	Not accepted by A	Total
Total	0.6	0.4	1.00

So the smallest the probability of being accepted at both schools can be is 0.3. This only occurs if you're guaranteed of getting into at least one of the two schools.

c. Your chances of getting into one school might change depending if you've gotten into the other. For example, your chances of getting to Cal Poly are probably higher if you know you've already gotten into Stanford.

d. For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55. If you are accepted at school A, what is your probability of also being accepted at school B?

	Accepted by A	Not accepted by A	Total
Accepted by B	0.55	0.15	0.70
Not accepted by B	0.05	0.25	0.30
Total	0.60	0.40	1.00

If you are accepted at school A, then your probability of also being accepted at school B is $\text{P}(B|A) = 0.55/0.60 = 0.917$. (91.7% of students like you who get into school A also get into school B.)

e. If you are accepted at school A, what is your probability of not being accepted at school B?

If you are accepted at school A, then your probability of not being accepted at school B is $\text{P}(B^c|A) = 0.05/0.60 = 0.083$. (8.3% of students like you who get into school A do not get into school B.)

f. If you are not accepted at school A, what is your probability of being accepted at school B?

If you are not accepted at school A, then your probability of not being accepted at school B is $\text{P}(B|A^c) = 0.15/0.40 = 0.375$. (37.5% of students like you who do not get into school A do get into school B.)

g. If you are accepted at school B, what is your probability of also being accepted at school A?

If you are accepted at school B, then your probability of being accepted at school A is $\text{P}(A|B) = 0.55/0.7 = 0.786$. (78.6% of students like you who get into school B also get into school A.)

h. If you are not accepted at school B, what is your probability of being accepted at school A?

   If you are not accepted at school B, then your probability of being accepted at school A is $\text{P}(A|B^c) = 0.05/30 = 0.167$. (16.7% of students like you who do not get into school B do get into school A.)

Problem 4

For each of the following, create your own "which of (a) or (b) is larger?" example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don't do any calculations.

a. Reverse the direction of conditioning (like the 6 foot tall and NBA example)
b. Compare conditional and unconditional probabilities (like man versus man who is greater than 6 feet tall)
c. Two "particulars" that people might think are not equally likely but are (like HHHHH versus HTHTT).
d. Comparing "the particular" and "the general" (HTHTT versus 2 H in 5 flips).
e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)

Solution

Results vary. See Section 1.6 of my probability textbook for discussion.

Problem 5

The "matching problem" involves n distincts objects labeled $1, \ldots, n$ which are placed in n distinct boxes labeled $1, \ldots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We're interested in the number of matches (the number of objects for which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

a. Describe in full detail how, in principle, you would physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $\text{P}(C)$.

c. This applet conducts a simulation of the matching problem (in the context of babies in a hospital being mixed up and returned to parents uniforomly at random.) In the applet, "Number of babies" represents n and "Number of trials" represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the "Randomize" button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $\mathrm{P}(C)$ when $n = 4$.
d. How do you expect $\mathrm{P}(C)$ to depend on n? As n increases, do you expect $\mathrm{P}(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you're free to choose other values). For each value of n run the simulation and use the results to approximate $\mathrm{P}(C)$. Record your results.
f. What do the simulation results suggest about the relationship between $\mathrm{P}(C)$ and n? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margion of error of each other are essentially the same approximation.)

Solution

a. Use a box with 4 tickets, labeled 1, 2, 3, 4. Shuffle the tickets and draw all 4 *without* replacement and record the tickets drawn *in order*. Let X be the number of tickets that match their spot in order. (For example, if the tickets are drawn in the order 2431 then the realized value of X is 1 since only ticket 3 matches its spot in the order.) Since $C = \{X \geq 1\}$, event C occurs if $X \geq 1$ and does not occur if $X = 0$. We could record the realization of event C as "True" or "False".
b. Repeat the previous part many times. Count the number of repetitions on which C occurs and divide by the total number of repetitions to approximate $\mathrm{P}(C)$.
c. See the applet.
d. You might think the probability might decrease with n, since the probability that any particular object goes in its correct spot decreases with n. On the other hand, as n increases, the more chances there are to get at least one object in the correct spot. These considerations move in opposite directions; what happens as n increases?
e. When $n = 4$, $\mathrm{P}(X = 0)$ is approximately 0.375; when $n = 5$, $\mathrm{P}(X = 0)$ is approximately 0.367; but for $n \geq 6$, $\mathrm{P}(X = 0)$ is approximately 0.368 for any value of n.
f. The probability of no matches essentially does not depend on n! We'll investigate a little more in class.

Problem 6

Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss's dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

a. Compute the probability that Katniss's dart lands within 1 inch of the center of the dartboard.
b. Compute the probability that Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Solution

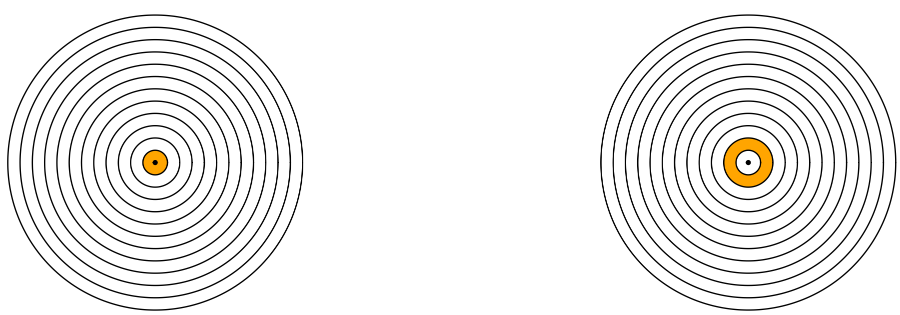
See Figure 18.1 for pictures.

1. Figure 18.1a: A, Katniss's dart lands within 1 inch of the center of the dartboard.
2. Figure 18.1b: B, Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Since the dart lands uniformly at random anywhere on the dartboard, probabilities are computed as the ratio between the area corresponding to the event of interest divided by the area of the total dartboard ($12^2\pi$ square-inches.)

Find the area of the shaded pieces by subtracting areas of circles.

1. $\mathrm{P}(A) = \frac{1^2\pi}{12^2\pi} = 1/144 = 0.00694$
2. $\mathrm{P}(B) = \frac{2^2\pi - 1^2\pi}{12^2\pi} = 3/144 = 0.021$



(a) Within 1 inch of center

(b) More than 1 but less than 2 inches from center

Figure 18.1: Events in dartboard problem

Problem 7

1. Imagine a light that flashes every few seconds[2]. The light randomly flashes green with probability 0.75 and red with probability 0.25, independently from flash to flash.

    a. Write down a sequence of G's (for green) and R's (for red) to predict the colors for the next 40 flashes of this light. Before you read on, please take a minute to think about how you would generate such a sequence yourself.
    b. Most people produce a sequence that has 30 G's and 10 R's, or close to those proportions, because they are trying to generate a sequence for which each outcome has a 75% chance for G and a 25% chance for R. That is, they use a strategy in which they predict G with probability 0.75, and R with probability 0.25. How well does this strategy do? Compute the probability of correctly predicting any single item in the sequence using this strategy.
    c. Describe a better strategy. (Hint: can you find a strategy for which the probability of correctly predicting any single flash is 0.75?)

Solution

a. Most people write a sequence with about 30 G and 10 R.
b. Use the law of total probability. There are two cases for your guesses: G or R.

    - If you guess G then you are correct if it really flashes G, and it flashes G with probability 0.75.
    - If you guess R then you are correct if it really flashes R, and it flashes R with probability 0.25.

    If you use a strategy where you guess G with probability 0.75, your probability of being correct is

$$\begin{aligned} \text{P(correct)} &= \text{P(correct}|\text{guess G)P(guess G)} + \text{P(correct}|\text{guess R)P(guess R)} \\ &= (0.75)(0.75) + (0.25)(0.25) = 0.625. \end{aligned}$$

c. If you get predict G every time, you are correct any time it is G, so the probability that you are correct is 0.75.

[2]Thanks to Allan Rossman for this example.

Summary of Some Common Distributions

Name and parameters	Possible values	pmf/pdf	cdf	Mean	SD
Discrete					
Poisson(μ)	$x = 0, 1, 2, \ldots$	$e^{-\mu}\frac{\mu^x}{x!}$	No closed form	μ	$\sqrt{\mu}$
Binomial(n, p)	$x = 0, 1, 2, \ldots, n$	$\binom{n}{x}p^x(1-p)^{n-x}$	No closed form	np	$\sqrt{np(1-p)}$
Geometric(p)	$x = 1, 2, 3, \ldots$	$p(1-p)^{x-1}$	$1-(1-p)^x$	$\frac{1}{p}$	$\sqrt{\frac{1-p}{p^2}}$
Continuous					
Uniform(a, b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{b-a}{\sqrt{12}}$
Exponential(λ)	$x > 0$	$\lambda e^{-\lambda x}$	$1-e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$
Normal(μ, σ)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$	No closed form (use tables)	μ	σ

References